```
# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py                启动并自动打开浏览器
    python run.py --no-browser   只启动服务，不打开浏览器
    python run.py --dev          开发模式（开启代码热重载与调试器）
    python run.py --port 5001    指定端口
默认以 127.0.0.1:5000 启动，就绪后自动打开浏览器。
"""
import os
import sys
import threading
import time
import webbrowser
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")   # 控制台中文正常显示
from app import create_app
from app.models import db
PORT = 5000
for i, a in enumerate(sys.argv):
    if a == "--port" and i + 1 < len(sys.argv):
        PORT = int(sys.argv[i + 1])
URL = "http://127.0.0.1:%d" % PORT
NO_BROWSER = "--no-browser" in sys.argv
DEV = "--dev" in sys.argv
app = create_app()
def open_browser_later(delay=1.2):
    """等服务工作起来再打开浏览器，避免打开时页面还没就绪。
    放在后台线程里，不用 sleep 阻塞服务启动；线程设为守护线程，
    用户 Ctrl+C 停止服务时不会残留。
    """
    def run():
        time.sleep(delay)
        try:
            webbrowser.open(URL)
        except Exception:
            pass            # 打开失败不影响服务本身
    t = threading.Thread(target=run, daemon=True)
    t.start()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()     # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)     # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("   历史遗留废弃矿山自然恢复适宜性评价系统  Ｖ１．０")
    print("=" * 58)
```

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    print("   访问地址：%s" % URL)
    print("   演示账号：admin      密码：admin123")
    print("   停止运行：按 Ctrl+C，或直接关闭本窗口")
    print("=" * 58)
    if not NO_BROWSER and not DEV:
        print("   正在打开浏览器……")
    print()
    # 开发模式才启用热重载（重载会二次启动进程，故此时不自动开浏览器）
    if not NO_BROWSER and not DEV:
        open_browser_later()
    app.run(host="127.0.0.1", port=PORT, debug=DEV,
    use_reloader=DEV)
# ==== app/__init__.py ====
# -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from datetime import date
from flask import Flask
from flask_login import LoginManager
from app.config import Config
from app.models import db, User
from app.utils.constants import ROLE_LABELS
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    # 注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report, search
    app.register_blueprint(auth.bp)
    app.register_blueprint(project.bp)
    app.register_blueprint(unit.bp)
    app.register_blueprint(standard.bp)
    app.register_blueprint(constraint.bp)
    app.register_blueprint(ahp.bp)
    app.register_blueprint(indicator.bp)
    app.register_blueprint(evaluate.bp)
    app.register_blueprint(report.bp)
    app.register_blueprint(search.bp)
    # 全局登录保护：除登录页与静态资源外，均需登录
    from flask import request, redirect, url_for,
    send_from_directory
    from flask_login import current_user, login_user
```

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PUBLIC_ENDPOINTS = {"auth.login", "auth.register",
"static", "favicon"}
#  浏览器默认会请求根路径下的 /favicon.ico。
#  页面里虽已用 <link rel="icon"> 指向 static 目录，但出错页等场景仍会
#  回退到这个路径，补一条路由免得日志里一直刷 404。
@app.route("/favicon.ico")
def favicon():
    return send_from_directory(app.static_folder,
"favicon.ico",
                               mimetype="image/vnd.micro
soft.icon")
@app.before_request
def require_login():
    #  截图模式（仅用于生成说明书配图）：自动以管理员身份登录
    if app.config.get("SCREENSHOT_MODE"):
        if (request.endpoint not in ("auth.login",
"auth.register", "static")
                and not current_user.is_authenticated):
            admin = User.query.filter_by(role="admin").f
irst()
            if admin:
                login_user(admin)
        return None
    if request.endpoint in PUBLIC_ENDPOINTS or
request.endpoint is None:
        return None
    if not current_user.is_authenticated:
        return redirect(url_for("auth.login"))
    return None
#  侧边栏导航：按论文评价流程的顺序编排，分为「评价流程」与「系统管理」两组。
#  不使用模块编号，避免出现与实际流程不一致的代号。
#  每项带图标名（内联 SVG，见 templates/_macros.html 的 ICONS）
#  与可选的待办角标（该步骤还没做完的图斑数）。
FLOW_LINKS = [
    ("项目与矿山档案", "/project/", "folder"),
    ("政策与标准知识库", "/standard/", "book"),
    ("限制性指标判定", "/constraint/", "shield"),
    ("微地质单元", "/unit/", "layers"),
    ("指标体系与权重", "/ahp/", "sliders"),
    ("指标数据采集", "/indicator/", "database"),
    ("适宜性定量评价", "/evaluate/", "target"),
    ("报告与结果输出", "/report/", "file"),
]
#  「用户与操作日志」已并入「个人设置」，不再是独立菜单，
#  因此系统管理组只留一个入口。
ADMIN_LINKS = [
    ("个人设置", "/profile", "user"),
]
@app.context_processor
def inject_nav():
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    path = request.path or "/"
    def hit(prefix):
        """当前路径是否属于某个入口（/unit/ 也匹配 /unit/3）。"""
        return path.startswith(prefix.rstrip("/"))
    badges = {}
    if current_user.is_authenticated:
        from app.services import dashboard as dash
        badges = dash.nav_badges()
    def build(links):
        return [{"label": label, "href": href, "icon": icon,
                 "active": hit(href), "badge": badges.get(href, 0)}
                for label, href, icon in links]
    groups = [
        {"title": "评价流程", "links": build(FLOW_LINKS)},
        {"title": "系统管理", "links": build(ADMIN_LINKS)},
    ]
    # 面包屑用的当前页名：按路径命中的入口取，未命中时给个通用名
    if path == "/":
        page_title = "工作台"
    elif path.startswith("/profile"):
        page_title = "个人设置"
    elif path.startswith(("/users", "/logs")):
        page_title = "账号与日志"
    elif path.startswith("/search"):
        page_title = "搜索结果"
    else:
        page_title = "工作台"
        for label, href, _icon in FLOW_LINKS:
            if hit(href):
                page_title = label
                break
    # 层级引导用：项目是归档单位，图斑才是评价对象。
    # 下游各页面按图斑组织数据，因此需要提示「哪些项目下还没有图斑」。
    from app.models import Project, MinePatch
    projects = Project.query.order_by(Project.id).all()
    return {
        "nav_groups": groups,
        "home_active": path == "/",
        "page_title": page_title,
        "today": date.today().strftime("%Y-%m-%d"),
        "search_q": request.args.get("q", ""),
        "role_labels": ROLE_LABELS,
        "project_count": len(projects),
        "patch_count": MinePatch.query.count(),
        "projects_without_patch": [p for p in projects
                                   if p.patches.count()
== 0],
    }
```

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    # 操作审计：统一在响应阶段记录成功的写操作（操作日志）
    from app.services import audit
    @app.after_request
    def audit_writes(response):
        if request.method in ("POST", "PUT", "PATCH",
    "DELETE"):
            audit.record(request.endpoint,
    response.status_code)
        return response
    # 错误页：给出可读提示，而不是默认的英文错误页
    from flask import render_template
    @app.errorhandler(403)
    def forbidden(e):
        return render_template("error.html", code=403,
                               message="没有权限执行该操作"), 403
    @app.errorhandler(404)
    def not_found(e):
        return render_template("error.html", code=404,
                               message="请求的页面不存在"), 404
    return app
# ==== app/config.py ====
# -*- coding: utf-8 -*-
"""应用配置。"""
import os
from datetime import timedelta
BASE_DIR = os.path.abspath(os.path.dirname(__file__))
class Config:
    SECRET_KEY = os.environ.get("SECRET_KEY",
    "mine-restoration-dev-key")
    # 单机部署，SQLite 数据库文件放在应用目录下
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE_DIR, "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项
    WTF_CSRF_ENABLED = True
    # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录
    REMEMBER_COOKIE_DURATION = timedelta(days=30)
    REMEMBER_COOKIE_HTTPONLY = True
    REMEMBER_COOKIE_SAMESITE = "Lax"
    # 影像与数据文件上传（遥感影像挂接、批量导入）
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限
    32 MB
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}
    # 头像只用浏览器能直接显示的格式（不含 bmp / tif），并单独限制体积
    ALLOWED_AVATAR_EXT = {"png", "jpg", "jpeg", "gif",
    "webp"}
```

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    MAX_AVATAR_LENGTH = 4 * 1024 * 1024          # 4 MB
# ==== app/models.py ====
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from datetime import datetime
from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（账户与权限）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
    nullable=False)
    # 职务只有两种：管理员 admin（可管理账号与他人的职务） / 操作员 operator
    role = db.Column(db.String(20), nullable=False,
    default="operator")
    active = db.Column(db.Boolean, nullable=False,
    default=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    # ---- 个人资料（用户可在「个人设置」里自行维护）----
    display_name = db.Column(db.String(64), nullable=True)
    # 昵称
    email = db.Column(db.String(128), nullable=True)
    phone = db.Column(db.String(32), nullable=True)
    gender = db.Column(db.String(8), nullable=True)
    # 男 / 女 / 保密
    avatar = db.Column(db.String(256), nullable=True)
    # 头像，相对 static 的路径
    @property
    def is_admin(self):
        return self.role == "admin"
    @property
    def shown_name(self):
        """界面上显示的名字：优先用昵称，没填就用账号名。"""
        return self.display_name or self.username
    @property
    def initial(self):
        """没有头像时用首字占位。"""
        return (self.shown_name or "?")[0].upper()
    def to_dict(self):
        return {"id": self.id, "username": self.username,
                "display_name": self.display_name,
```

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        "gender": self.gender,                                          301
                    "role": self.role, "active": self.active}           302
class OperationLog(db.Model):                                           303
    """操作日志。"""                                                    304
    __tablename__ = "operation_logs"                                    305
    id = db.Column(db.Integer, primary_key=True)                        306
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"            307
    ), nullable=True)                                                   308
    action = db.Column(db.String(128), nullable=False)                  309
    # 操作类型                                                          310
    target_type = db.Column(db.String(64), nullable=True)               311
    # 对象类型                                                          312
    target_id = db.Column(db.Integer, nullable=True)                    313
    # 对象主键                                                          314
    detail = db.Column(db.Text, nullable=True)                          315
    # 操作详情                                                          316
    created_at = db.Column(db.DateTime,                                 317
    default=datetime.utcnow, index=True)                                318
class Project(db.Model):                                                319
    """评价项目（项目与矿山档案）。"""                                  320
    __tablename__ = "projects"                                          321
    id = db.Column(db.Integer, primary_key=True)                        322
    name = db.Column(db.String(128), nullable=False)                    323
    code = db.Column(db.String(64), nullable=True)                      324
    # 项目编号                                                          325
    description = db.Column(db.Text, nullable=True)                     326
    status = db.Column(db.String(20), default="进行中")                 327
    # 进行中 / 已完成 / 已归档                                          328
    created_by = db.Column(db.Integer,                                  329
    db.ForeignKey("users.id"), nullable=True)                           330
    created_at = db.Column(db.DateTime,                                 331
    default=datetime.utcnow)                                            332
    patches = db.relationship("MinePatch",                              333
    backref="project", lazy="dynamic",                                  334
                                    cascade="all, delete-orphan")       335
class MinePatch(db.Model):                                              336
    """矿山图斑。"""                                                    337
    __tablename__ = "mine_patches"                                      338
    id = db.Column(db.Integer, primary_key=True)                        339
    project_id = db.Column(db.Integer,                                  340
    db.ForeignKey("projects.id"), nullable=False)                       341
    name = db.Column(db.String(128), nullable=False)                    342
    admin_region = db.Column(db.String(128), nullable=True)             343
    # 所在行政区                                                        344
    center_x = db.Column(db.Float, nullable=True)                       345
    # 中心经度 / 横坐标                                                 346
    center_y = db.Column(db.Float, nullable=True)                       347
    # 中心纬度 / 纵坐标                                                 348
    area_ha = db.Column(db.Float, nullable=True)                        349
    # 图斑面积（公顷）                                                  350
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    mineral_type = db.Column(db.String(64), nullable=True)
    # 开采矿种
    mining_method = db.Column(db.String(64), nullable=True)
    # 开采方式
    stop_years = db.Column(db.Float, nullable=True)
    # 停产年限
    responsible_entity_missing = db.Column(db.Boolean,
    default=True)  # 责任主体是否缺失
    # 遥感底图 / 无人机正射影像挂接
    image_path = db.Column(db.String(256), nullable=True)
    # 相对 static 的路径
    image_note = db.Column(db.String(256), nullable=True)
    # 影像说明（如来源、时相）
    image_uploaded_at = db.Column(db.DateTime,
    nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    units = db.relationship("MicroUnit", backref="patch",
    lazy="dynamic",
                            cascade="all, delete-orphan")
class ReferenceEcosystem(db.Model):
    """参照生态系统：标准化基准值，与图斑绑定。"""
    __tablename__ = "reference_ecosystems"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    coverage_benchmark = db.Column(db.Float, default=75.0)
    # 植被覆盖度基准 %
    shannon_benchmark = db.Column(db.Float, default=0.7)
    # Shannon-Wiener 基准
    dominant_species = db.Column(db.String(256),
    nullable=True)  # 建群种
    soil_benchmark = db.Column(db.String(256),
    nullable=True)  # 土壤基准
class Standard(db.Model):
    """地方标准 / 政策文件（政策与标准知识库）。"""
    __tablename__ = "standards"
    id = db.Column(db.Integer, primary_key=True)
    province = db.Column(db.String(32), nullable=False)
    # 省份
    name = db.Column(db.String(256), nullable=False)
    # 名称
    doc_number = db.Column(db.String(128), nullable=True)
    # 标准编号 / 文号
    publish_date = db.Column(db.String(32), nullable=True)
    implement_date = db.Column(db.String(32), nullable=True)
    # 类别：地方标准 / 政策文件
    category = db.Column(db.String(32), nullable=False,
    default="地方标准")
    summary = db.Column(db.Text, nullable=True)
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    # 特点概述
    rules = db.relationship("StandardRule",
    backref="standard",
                            cascade="all, delete-orphan")
class StandardRule(db.Model):
    """结构化规则：将标准条文拆解为可执行规则。"""
    __tablename__ = "standard_rules"
    id = db.Column(db.Integer, primary_key=True)
    standard_id = db.Column(db.Integer,
    db.ForeignKey("standards.id"), nullable=False)
    item = db.Column(db.String(128), nullable=False)
    # 判定项
    operator = db.Column(db.String(8), nullable=False)
    # 比较符：> >= < <= ==
    threshold = db.Column(db.String(64), nullable=True)
    # 阈值
    condition_type = db.Column(db.String(32),
    default="必要条件")  # 必要条件 / 充分条件
    region = db.Column(db.String(64), nullable=True)
    # 适用分区（宁夏分区差异化）
    note = db.Column(db.Text, nullable=True)
class ConstraintCheck(db.Model):
    """限制性判定记录。"""
    __tablename__ = "constraint_checks"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    policy_pass = db.Column(db.Boolean, nullable=True)
    # 政策约束
    location_pass = db.Column(db.Boolean, nullable=True)
    # 区位约束
    pollution_pass = db.Column(db.Boolean, nullable=True)
    # 土壤污染
    geo_pass = db.Column(db.Boolean, nullable=True)
    # 地质灾害
    farmland_pass = db.Column(db.Boolean, nullable=True)
    # 耕地属性
    passed = db.Column(db.Boolean, nullable=True)
    # 总体是否通过
    reason = db.Column(db.Text, nullable=True)
    # 不通过原因
    checked_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class MicroUnit(db.Model):
    """微地质单元：岩质边坡 / 碎石边坡 / 平台。"""
    __tablename__ = "micro_units"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    unit_type = db.Column(db.String(32), nullable=False)
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    # 单元类型
    area = db.Column(db.Float, nullable=True)
    # 面积
    ratio = db.Column(db.Float, nullable=True)
    # 面积占比（自动计算）
    description = db.Column(db.Text, nullable=True)
    # 现场描述
class Indicator(db.Model):
    """指标定义：指标树，含角色与启用单元类型。"""
    __tablename__ = "indicators"
    id = db.Column(db.Integer, primary_key=True)
    code = db.Column(db.String(32), unique=True,
    nullable=False)  # 指标编码
    name = db.Column(db.String(64), nullable=False)
    # 指标名称
    level = db.Column(db.Integer, nullable=False)
    # 1/2/3 级
    parent_code = db.Column(db.String(32), nullable=True)
    # 父指标编码
    # 角色：土壤污染指标同时作为限制性指标与评价指标
    is_constraint = db.Column(db.Boolean, default=False)
    is_evaluative = db.Column(db.Boolean, default=True)
    unit = db.Column(db.String(32), nullable=True)
    # 单位
    ideal_value = db.Column(db.String(64), nullable=True)
    # 理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    # benefit/cost/mid/discrete
    # 启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
    # 取值依据说明
    def units_list(self):
        return [s for s in (self.enabled_units or
    "").split(",") if s]
class WeightScheme(db.Model):
    """权重方案：按微地质单元类型分别保存。"""
    __tablename__ = "weight_schemes"
    id = db.Column(db.Integer, primary_key=True)
    unit_type = db.Column(db.String(32), nullable=False)
    # 所属单元类型
    name = db.Column(db.String(64), nullable=True)
    # 方案名称
    level = db.Column(db.Integer, nullable=False,
    default=1)        # 1=一级指标 2=二级指标
    is_default = db.Column(db.Boolean, default=False)
    # 是否默认（论文表 3）
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class WeightItem(db.Model):
```

```
    """权重明细：方案内的各项权重。"""                                    501
    __tablename__ = "weight_items"                                      502
    id = db.Column(db.Integer, primary_key=True)                        503
    scheme_id = db.Column(db.Integer,                                   504
    db.ForeignKey("weight_schemes.id"), nullable=False)                 505
    indicator_code = db.Column(db.String(32),                           506
    nullable=False)       # 指标编码                                    507
    weight = db.Column(db.Float, nullable=False)                        508
    # 权重值                                                            509
class ExpertMatrix(db.Model):                                           510
    """专家判断矩阵：多专家打分的原始矩阵。"""                            511
    __tablename__ = "expert_matrices"                                   512
    id = db.Column(db.Integer, primary_key=True)                        513
    scheme_id = db.Column(db.Integer,                                   514
    db.ForeignKey("weight_schemes.id"), nullable=True)                  515
    # 直接标注适用对象，便于无需先建权重方案即可录入专家矩阵             516
    unit_type = db.Column(db.String(32), nullable=True)                 517
    # 微地质单元类型                                                    518
    l1_name = db.Column(db.String(64), nullable=True)                   519
    # 所属一级指标（二级矩阵用）                                        520
    expert_name = db.Column(db.String(64), nullable=True)               521
    # 专家姓名                                                          522
    level = db.Column(db.Integer, nullable=False,                       523
    default=1)          # 矩阵对应指标层级                              524
    matrix_json = db.Column(db.Text, nullable=False)                    525
    # 判断矩阵（JSON 二维数组）                                         526
    cr = db.Column(db.Float, nullable=True)                             527
    # 该专家矩阵的一致性比例                                            528
    created_at = db.Column(db.DateTime,                                 529
    default=datetime.utcnow)                                            530
class IndicatorData(db.Model):                                          531
    """指标实测数据：按微地质单元组织。"""                               532
    __tablename__ = "indicator_data"                                    533
    id = db.Column(db.Integer, primary_key=True)                        534
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni            535
    ts.id"), nullable=False, unique=True)                               536
    soil_ratio = db.Column(db.Float, nullable=True)                     537
    # 土石比                                                            538
    thickness = db.Column(db.Float, nullable=True)                      539
    # 土层厚度 cm                                                       540
    ph = db.Column(db.Float, nullable=True)                             541
    organic_matter = db.Column(db.Float, nullable=True)                 542
    # 有机质 g/kg                                                       543
    # 重金属含量，JSON 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null   544
    heavy_metals = db.Column(db.Text, nullable=True)                    545
    coverage = db.Column(db.Float, nullable=True)                       546
    # 植被覆盖度 %                                                      547
    shannon = db.Column(db.Float, nullable=True)                        548
    # Shannon-Wiener 多样性指数                                         549
    slope = db.Column(db.Float, nullable=True)                          550
```

```
    # 坡度 °
    aspect = db.Column(db.String(16), nullable=True)
    # 坡向：阳坡 / 阴阳坡 / 阴坡
    def heavy_metals_dict(self):
        return json.loads(self.heavy_metals) if
    self.heavy_metals else {}
class EvaluationResult(db.Model):
    """评价结果。"""
    __tablename__ = "evaluation_results"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    final_score = db.Column(db.Float, nullable=True)
    # 最终评价因子 F
    grade = db.Column(db.String(32), nullable=True)
    # 适宜性等级
    measures = db.Column(db.Text, nullable=True)
    # 推荐措施
    unit_scores = db.Column(db.Text, nullable=True)
    # 各单元得分 JSON
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class ScoreDetail(db.Model):
    """得分明细：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
import re
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
```

```
from wtforms.validators import (DataRequired, Length,
    Optional, EqualTo,
                                ValidationError)
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES,
    GENDERS
#  邮箱格式：够用的轻量校验。
#  WTForms 自带的 Email() 需要额外安装 email_validator,
#  为一个字段引入依赖不划算，这里自己用正则判断。
EMAIL_RE = re.compile(r"^[^@\s]+@[^@\s]+\.[A-Za-z]{2,}$")
def email_format(form, field):
    v = (field.data or "").strip()
    if v and not EMAIL_RE.match(v):
        raise ValidationError("邮箱格式不正确")
def email_ok(value):
    """邮箱格式判断，供不用 WTForms 的表单复用（如管理员编辑用户）。"""
    v = (value or "").strip()
    return (not v) or bool(EMAIL_RE.match(v))
class AHPSelectForm(FlaskForm):
    """选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  #  动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
        self.unit_type.choices = [(t, t) for t in
    UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """登录表单。支持「记住我」以在关闭浏览器后保持登录状态。"""
    username = StringField("账户名称", validators=[DataRequired(
    ), Length(max=64)])
    password = PasswordField("密码", validators=[DataRequired(
    )])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
GENDER_CHOICES = [("", "不填")] + [(g, g) for g in GENDERS]
class ProfileForm(FlaskForm):
    """个人资料：昵称、性别与联系方式。字段都可留空。
    职务不在这里 —— 只能由管理员在「个人设置 · 账号」里改。
    """
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
```

```
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_info = SubmitField("保存资料")
class RegisterForm(FlaskForm):
    """注册：账号与密码必填，其余选填。注册后默认是操作员。"""
    username = StringField("账号名称",
                           validators=[DataRequired("请填写账号名称
    "),
                                       Length(min=2, max=64,
                                              message="账号名称
    2–64 个字符")])
    password = PasswordField("密码",
                             validators=[DataRequired("请设置密码
    "),
                                         Length(min=6,
    max=64,

    message="密码至少 6 位")])
    confirm = PasswordField("确认密码",
                            validators=[DataRequired(),
                                        EqualTo("password",

    message="两次输入的密码不一致")])
    display_name = StringField("昵称", validators=[Optional(),
    Length(max=64)])
    gender = SelectField("性别", choices=GENDER_CHOICES,
    validators=[Optional()])
    email = StringField("邮箱", validators=[Optional(),
    email_format,
                                            Length(max=128)])
    phone = StringField("手机号", validators=[Optional(),
    Length(max=32)])
    submit_reg = SubmitField("注册并登录")
class PasswordForm(FlaskForm):
    """修改密码：需要先验证当前密码。"""
    old_password = PasswordField("当前密码",
    validators=[DataRequired()])
    new_password = PasswordField("新密码",
                                 validators=[DataRequired(),
    Length(min=6, max=64,

    message="新密码至少 6 位")])
    confirm = PasswordField("确认新密码",
                            validators=[DataRequired(),
```

```
        EqualTo("new_password",
        message="两次输入的新密码不一致")])
        submit_pwd = SubmitField("修改密码")
# ==== app/utils/constants.py ====
# -*- coding: utf-8 -*-
"""评价指标常量表。
全部数值取自论文，作为系统计算的唯一基准：
- 论文 4.2.2 节：指标标准化理想极值与取值依据
- 论文 4.2.3 节 / 表 4：适宜性等级划分
- 论文表 3：按微地质单元分别确定的一级、二级指标权重
- 论文 4.2.1 节：平均随机一致性指标 RI
"""
# ---------------------------------------------------------------------
    ------------------
# 一、微地质单元类型
# ---------------------------------------------------------------------
    ------------------
# 论文 4.1 节："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡
# 三种地质微单元类型"
UNIT_TYPES = ("岩质边坡", "碎石边坡", "平台")
# 每个单元类型启用的指标类别（论文表 3 隐含：岩质边坡无土壤指标，平台无地形指标）
# 供数据录入界面与评价引擎判断"该单元是否需要某类指标"
UNIT_ENABLED_INDICATORS = {
    "岩质边坡": ("植被指标", "地形指标"),                    # 不设土壤指标
    "碎石边坡": ("土壤指标", "植被指标", "地形指标"),
    "平台": ("土壤指标", "植被指标"),                        # 不设地形指标
}
# ---------------------------------------------------------------------
    ------------------
# 二、指标标准化理想极值（论文 4.2.2 节）
# ---------------------------------------------------------------------
    ------------------
IDEAL_SOIL_RATIO = 0.8          # 土石比理想值
IDEAL_THICKNESS = 20.0          # 土层厚度理想值 （cm）
IDEAL_PH_RANGE = (5.0, 9.0)     # pH 理想区间（适中型指标）
IDEAL_ORGANIC = 30.0            # 有机质理想值
    （g/kg），全国第二次土壤普查养分等级划分标准二等最低值
IDEAL_COVERAGE = 75.0           # 植被覆盖度参照生态系统 （%）
IDEAL_SHANNON = 0.7             # Shannon-Wiener 多样性指数参照生态系统
# 重金属极值 （mg/kg），《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值
# 论文原文：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、530、400、45、220、600、800
HEAVY_METALS = ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu",
    "Zn")
HEAVY_METAL_LIMIT = {
    "Cd": 2.0, "Hg": 2.0, "Pb": 530.0, "Cr": 400.0,
    "As": 45.0, "Ni": 220.0, "Cu": 600.0, "Zn": 800.0,
}
# 污染指标下 8 种重金属"重要性相当，权重平均分配"（论文 4.2.1 节）
HEAVY_METAL_EQUAL_WEIGHT = 1.0 / 8.0
```

```
#  坡度分界，论文  4．2．2  节引《新一轮退耕还林还草总体方案》发改西部〔２０１４〕１７７２  号
SLOPE_BREAK  =  25．0
#  坡向离散赋值，论文  4．2．2  节："阳坡、阴阳坡、阴坡分别赋值  0．2、0．6、1"
ASPECT_SCORES  =  {"阳坡":  0．2，  "阴阳坡":  0．6，  "阴坡":  1．0}
ASPECT_TYPES  =  ("阳坡",  "阴阳坡",  "阴坡")
#  pH  适中型标准化的容差边界（区间外线性衰减，避免得分突变）
PH_TOL_LOW  =  4．0
PH_TOL_HIGH  =  10．0
#  ----------------------------------------------------------------
      ------------------
#  三、适宜性等级划分（论文表  4）
#  ----------------------------------------------------------------
      ------------------
GRADE_LEVELS  =  (
      {"grade":  "不适宜",  "min":  None,  "max":  0．3，
        "meaning":  "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复"，
        "measure":  "生态重建、人工辅助"}，
      {"grade":  "基本适宜",  "min":  0．3，  "max":  0．7，
        "meaning":  "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求"，
        "measure":  "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"}，
      {"grade":  "适宜",  "min":  0．7，  "max":  None，
        "meaning":  "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快"，
        "measure":  "自然恢复"}，
)
def  grade_of(score):
      """按论文表  4  判定适宜性等级。"""
      for  g  in  GRADE_LEVELS:
            lo，  hi  =  g["min"]，  g["max"]
            if  lo  is  None  and  score  <  hi:
                  return  g
            if  hi  is  None  and  score  >=  lo:
                  return  g
            if  lo  is  not  None  and  hi  is  not  None  and  lo  <=
      score  <  hi:
                  return  g
      #  score  恰好等于  0．7  时落入"适宜"
      return  GRADE_LEVELS[-1]
#  ----------------------------------------------------------------
      ------------------
#  四、平均随机一致性指标  RI（论文  4．2．1  节给出  n=2，3，4，5，其余为常用扩展值）
#  ----------------------------------------------------------------
      ------------------
RI_TABLE  =  {1:  0．0，  2:  0．0，  3:  0．58，  4:  0．90，  5:  1．12，  6:
      1．24，
                  7:  1．32，  8:  1．41，  9:  1．45，  10:  1．49}
#  一致性比例阈值（论文：当  CR<0．1  时，判断矩阵满足一致性要求）
CR_THRESHOLD  =  0．1
#  ----------------------------------------------------------------
      ------------------
#  五、默认权重方案（论文表  3）
```

```
#  ----------------------------------------------------------------------  801
    ----------------                                                        802
#  论文　４．２．２　节："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，  803
#  分别计算评价指标权重"（论文摘要、图　２　流程与表　５　均作"碎石边坡"，  804
#  系统统一采用"碎石边坡"）。权重由　１０　位专家打分后经　ＡＨＰ　方法确定。  805
#  下表为论文表　３　所载的归一化权重，作为系统默认方案。                  806
DEFAULT_WEIGHTS　=　{                                                      807
    "岩质边坡"：　{                                                        808
        "ｌ１"：　{"植被指标"：　０．４５，　"地形指标"：　０．５５},        809
        "ｌ２"：　{                                                        810
            "植被指标"：　{"植被覆盖度"：　０．６７，　"生物多样性"：　０．３３},  811
            "地形指标"：　{"坡度"：　０．６４，　"坡向"：　０．３６},       812
        },                                                                 813
    },                                                                     814
    "碎石边坡"：　{                                                        815
        "ｌ１"：　{"土壤指标"：　０．４４，　"植被指标"：　０．３１，　"地形指标"：　０．２５},  816
        "ｌ２"：　{                                                        817
            "土壤指标"：　{"土石比"：　０．２５，　"土层厚度"：　０．２５，　"ｐＨ"：　０．１７，  818
                          "有机质"：　０．１７，　"污染指标"：　０．１７},  819
            "植被指标"：　{"植被覆盖度"：　０．６４，　"生物多样性"：　０．３６},  820
            "地形指标"：　{"坡度"：　０．６７，　"坡向"：　０．３３},       821
        },                                                                 822
    },                                                                     823
    "平台"：　{                                                            824
        "ｌ１"：　{"土壤指标"：　０．５４，　"植被指标"：　０．４６},        825
        "ｌ２"：　{                                                        826
            "土壤指标"：　{"土石比"：　０．２５，　"土层厚度"：　０．２５，　"ｐＨ"：　０．１７，  827
                          "有机质"：　０．１７，　"污染指标"：　０．１７},  828
            "植被指标"：　{"植被覆盖度"：　０．６１，　"生物多样性"：　０．３９},  829
        },                                                                 830
    },                                                                     831
}                                                                          832
#  ----------------------------------------------------------------------  833
    ----------------                                                        834
#  六、限制性指标（论文表　２　+　４．１　节流程）                          835
#  ----------------------------------------------------------------------  836
    ----------------                                                        837
#  论文摘要与　４．１　节：四项限制性指标；流程段另明确"是否为耕地"为约束条件  838
CONSTRAINT_ITEMS　=　("政策"，　"区位约束"，　"土壤污染"，　"地质灾害"，　"耕地属性")  839
#  ----------------------------------------------------------------------  840
    ----------------                                                        841
#  七、账号职务                                                            842
#  ----------------------------------------------------------------------  843
    ----------------                                                        844
#  只分两种：管理员可管理账号、维护基础数据并查看操作日志，操作员用业务功能。  845
ROLE_LABELS　=　{"admin"：　"管理员"，　"operator"：　"操作员"}            846
ROLE_CODES　=　("admin"，　"operator")                                     847
DEFAULT_ROLE　=　"operator"　　#　新注册账号的默认职务                    848
ROLE_ORDER　=　("admin"，　"operator")　　#　列表排序用：管理员在前        849
#  性别（注册与个人资料中选填）                                            850
```

```
GENDERS = ("男", "女", "保密")
# ==== app/utils/schema.py ====
# -*- coding: utf-8 -*-
"""轻量结构迁移。
SQLite 下 `db.create_all()` 只建缺失的表，不会给已有表补列。
本模块在启动时检查目标列是否存在，缺失则 ALTER TABLE 补上，
使旧数据库无需重建即可用上新字段。
"""
from sqlalchemy import text
# （表名， 列名， 列定义）
REQUIRED_COLUMNS = [
    # 遥感影像挂接
    ("mine_patches", "image_path", "VARCHAR(256)"),
    ("mine_patches", "image_note", "VARCHAR(256)"),
    ("mine_patches", "image_uploaded_at", "DATETIME"),
    # 指标维护
    ("indicators", "enabled_units", "VARCHAR(128)"),
    ("indicators", "description", "TEXT"),
    # 多专家矩阵
    ("expert_matrices", "unit_type", "VARCHAR(32)"),
    ("expert_matrices", "l1_name", "VARCHAR(64)"),
    ("expert_matrices", "cr", "FLOAT"),
    # 用户个人资料（账户页）
    ("users", "display_name", "VARCHAR(64)"),
    ("users", "email", "VARCHAR(128)"),
    ("users", "phone", "VARCHAR(32)"),
    ("users", "gender", "VARCHAR(8)"),
    ("users", "avatar", "VARCHAR(256)"),
]
# 职务体系由「管理员 / 评价员 / 查看者」三种收敛为「管理员 / 操作员」两种，
# 旧数据里的后两种统一归为操作员。
ROLE_MIGRATION = {"evaluator": "operator", "viewer":
    "operator"}
def _add_columns(db):
    added = []
    with db.engine.connect() as conn:
        for table, column, ddl in REQUIRED_COLUMNS:
            rows = conn.execute(text("PRAGMA
    table_info(%s)" % table)).fetchall()
            if not rows:
                continue                                #
    表还不存在，create_all 会建
            existing = {r[1] for r in rows}
            if column in existing:
                continue
            conn.execute(text(
                "ALTER TABLE %s ADD COLUMN %s %s" % (table,
    column, ddl)))
            conn.commit()
            added.append("%s.%s" % (table, column))
```

```
    return added                                                    901
def _migrate_roles(db):                                             902
    """把旧角色名统一为新的两种，返回改动的行数。"""                    903
    changed = 0                                                     904
    with db.engine.connect() as conn:                               905
        rows = conn.execute(text("PRAGMA                            906
    table_info(users)")).fetchall()                                 907
        if not rows:                                                908
            return 0                                                909
        for old, new in ROLE_MIGRATION.items():                     910
            r = conn.execute(text("UPDATE users SET role =          911
    :new "                                                          912
                                  "WHERE role = :old"),             913
                                 {"new": new, "old": old})          914
            changed += r.rowcount or 0                              915
        conn.commit()                                               916
    return changed                                                  917
def ensure_schema(db):                                              918
    """为已有表补齐缺失列，并做必要的数据迁移。返回变更清单。"""        919
    try:                                                            920
        changed = _add_columns(db)                                  921
        n = _migrate_roles(db)                                      922
        if n:                                                       923
            changed.append("用户角色归一为管理员/操作员 %d 条" % n)  924
        return changed                                              925
    except Exception as e:                                  #       926
    迁移失败不应阻断启动                                            927
        print("[schema] 结构检查跳过：%s" % e)                      928
        return []                                                   929
# ==== app/utils/uploads.py ====                                    930
# -*- coding: utf-8 -*-                                             931
"""上传文件处理（遥感影像、批量导入）。"""                           932
import os                                                           933
import re                                                           934
import uuid                                                         935
from datetime import datetime                                       936
from werkzeug.utils import secure_filename                          937
def _safe_ext(filename):                                            938
    return os.path.splitext(filename)[1].lower().lstrip(".")        939
def ext_allowed(filename, allowed):                                 940
    return _safe_ext(filename) in allowed                           941
def save_upload(file_storage, subdir, allowed_ext,                  942
    upload_root):                                                   943
    """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。  944
    文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。      945
    """                                                             946
    if file_storage is None or not file_storage.filename:           947
        return None, "未选择文件"                                   948
    ext = _safe_ext(file_storage.filename)                          949
    if ext not in allowed_ext:                                      950
```

```
            return None, "不支持的文件类型：.%s" % ext                    951
        target_dir = os.path.join(upload_root, subdir)                    952
        os.makedirs(target_dir, exist_ok=True)                            953
        # 仅保留安全字符，并加时间戳与随机串防重名                          954
        base = secure_filename(os.path.splitext(file_storage.fil          955
        ename)[0]) or "file"                                              956
        base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or               957
        "file"                                                            958
        name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M          959
        %S"),                                                             960
                                    uuid.uuid4().hex[:6], ext)            961
        abs_path = os.path.join(target_dir, name)                         962
        file_storage.save(abs_path)                                       963
        rel = os.path.join("uploads", subdir,                             964
        name).replace("\\", "/")                                          965
        return rel, None                                                  966
def remove_upload(rel_path, upload_root):                                 967
        """删除已保存的上传文件（相对 static 的路径）。"""                 968
        if not rel_path:                                                  969
            return                                                        970
        abs_path = os.path.join(upload_root,                              971
        rel_path.replace("uploads/", "", 1))                              972
        try:                                                              973
            if os.path.isfile(abs_path):                                  974
                os.remove(abs_path)                                       975
        except OSError:                                                   976
            pass                                                          977
# ==== app/utils/charts.py ====                                           978
# -*- coding: utf-8 -*-                                                   979
"""轻量图表。                                                             980
用纯 SVG 输出，不依赖任何绘图库或前端脚本：                               981
- 服务端渲染，页面与打印（PDF）中都能稳定显示                             982
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）               983
- 单色方案，符合界面「一个视图一个强调色」的约定                          984
"""                                                                       985
from html import escape                                                   986
ACCENT = "#1f3a5f"                                                        987
ACCENT_SOFT = "#7f93ad"                                                   988
INK = "#23262b"                                                           989
MUTED = "#6b7280"                                                         990
LINE = "#dfe3e8"                                                          991
# 灰度梯度：黑白打印时相邻扇区 / 条柱仍可区分                            992
GRAYS = ["#1f3a5f", "#456188", "#6d86a8", "#96a9c4",                      993
        "#bcc9da", "#dbe3ec"]                                             994
def _fmt(v, digits=4):                                                    995
        try:                                                              996
            return ("%%.%df" % digits) % float(v)                         997
        except (TypeError, ValueError):                                   998
            return str(v)                                                 999
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,               1000
```

```
    value_w=76):                                                    1001
    """水平条形图。items: [(标签, 数值, 显示文本)]"""                    1002
    if not items:                                                   1003
        return '<p class="muted">暂无数据</p>'                       1004
    n = len(items)                                                  1005
    height = n * (bar_h + gap) + 10                                 1006
    max_v = max((abs(float(v)) for _, v, _ in items),               1007
    default=1) or 1                                                 1008
    body_w = width - label_w - value_w                              1009
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"                1010
    height="%d" '                                                   1011
             'role="img" aria-label="条形图" '                       1012
             'style="max-width:%dpx;font-family:inherit;">'         1013
    % (width, height, height, width)]                               1014
    for i, (label, value, disp) in enumerate(items):                1015
        y = i * (bar_h + gap) + 5                                   1016
        frac = (abs(float(value)) / max_v) if max_v else 0          1017
        w = max(2.0, body_w * frac)                                 1018
        fill = GRAYS[i % len(GRAYS)]                                1019
        parts.append(                                               1020
            '<text x="%d" y="%d" text-anchor="end"                  1021
    font-size="12" fill="%s">%s</text>'                             1022
            % (label_w - 10, y + bar_h * 0.72, INK,                 1023
    escape(str(label))))                                            1024
        parts.append(                                               1025
            '<rect x="%d" y="%d" width="%.1f" height="%d"           1026
    rx="3" fill="%s"/>'                                             1027
            % (label_w, y, w, bar_h, fill))                         1028
        parts.append(                                               1029
            '<text x="%d" y="%d" font-size="12" fill="%s" '         1030
            'style="font-variant-numeric:tabular-nums;">%s</        1031
    text>'                                                          1032
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,            1033
    escape(str(disp))))                                             1034
    parts.append('</svg>')                                          1035
    return "".join(parts)                                           1036
def donut(items, size=210, thickness=34):                           1037
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]              1038
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。                1039
    """                                                             1040
    vals = [(l, float(v), d) for l, v, d in items if                1041
    float(v) > 0]                                                   1042
    if not vals:                                                    1043
        return '<p class="muted">暂无数据</p>'                       1044
    total = sum(v for _, v, _ in vals) or 1                         1045
    r = (size - thickness) / 2                                      1046
    cx = cy = size / 2                                              1047
    import math                                                     1048
    parts = ['<svg viewBox="0 0 %d %d" width="%d"                   1049
    height="%d" role="img" '                                        1050
```

```
            'aria-label="权重分布环形图">' % (size, size, size,
    size)]
    angle = -math.pi / 2
    for i, (label, v, disp) in enumerate(vals):
        sweep = 2 * math.pi * (v / total)
        a0, a1 = angle, angle + sweep
        x0, y0 = cx + r * math.cos(a0), cy + r *
    math.sin(a0)
        x1, y1 = cx + r * math.cos(a1), cy + r *
    math.sin(a1)
        large = 1 if sweep > math.pi else 0
        parts.append(
            '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f
    %.2f" '
            'fill="none" stroke="%s" stroke-width="%d"/>'
           % (x0, y0, r, r, large, x1, y1, GRAYS[i %
    len(GRAYS)], thickness))
        angle = a1
    parts.append('<text x="%d" y="%d" text-anchor="middle"
    font-size="12" '
                 'fill="%s">合计</text>' % (cx, cy - 2,
    MUTED))
    parts.append('<text x="%d" y="%d" text-anchor="middle"
    font-size="15" '
                 'fill="%s" style="font-variant-numeric:tabu
    lar-nums;">%s</text>'
                 % (cx, cy + 18, INK, _fmt(total, 4)))
    parts.append('</svg>')
    legend = ['<table style="width:auto;margin-top:10px;"><t
    head><tr>'
              '<th scope="col">项目</th><th
    scope="col">数值</th>'
              '<th scope="col">占比</th></tr></thead><tbody>']
    for i, (label, v, disp) in enumerate(vals):
        legend.append(
            '<tr><td style="text-align:left;">'
            '<span style="display:inline-block;width:10px;he
    ight:10px;'
            'border-radius:2px;background:%s;margin-right:6p
    x;"></span>%s</td>'
            '<td class="mono">%s</td><td
    class="mono">%.1f%%</td></tr>'
           % (GRAYS[i % len(GRAYS)], escape(str(label)),
    escape(str(disp)),
              100.0 * v / total))
    legend.append('</tbody></table>')
    return ('<div style="display:flex;flex-wrap:wrap;gap:20p
    x;align-items:center;">'
            + "".join(parts) + "".join(legend) + '</div>')
# ==== app/utils/browser.py ====
```

```
# -*- coding: utf-8 -*-
"""HTML 转 PDF（报告导出）。
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，
避免中文路径在 file:// URL 下出问题。
"""
import os
import shutil
import subprocess
import tempfile
import uuid
CANDIDATES = [
    r"C:\Program Files\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe",
    r"C:\Program Files\Microsoft\Edge\Application\msedge.exe",
]
# 允许通过环境变量覆盖，便于在其他机器上指定浏览器
ENV_KEY = "DSH_BROWSER_PATH"
def browser_path():
    """定位可用的浏览器可执行文件，找不到返回 None。"""
    override = os.environ.get(ENV_KEY)
    if override and os.path.isfile(override):
        return override
    for p in CANDIDATES:
        if os.path.isfile(p):
            return p
    for name in ("chrome", "msedge", "chromium"):
        found = shutil.which(name)
        if found:
            return found
    return None
def html_to_pdf(html, timeout=90):
    """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""
    exe = browser_path()
    if exe is None:
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")
    tag = uuid.uuid4().hex[:8]
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)
    try:
        with open(html_path, "w", encoding="utf-8") as f:
            f.write(html)
        cmd = [
```

```
                exe, "--headless=new", "--disable-gpu",
    "--no-sandbox",
                "--no-first-run", "--no-default-browser-check",
                "--no-pdf-header-footer",
                "--virtual-time-budget=4000",
                "--print-to-pdf=%s" % pdf_path,
                "file:///" + html_path.replace("\\", "/"),
            ]
            p = subprocess.run(cmd, capture_output=True,
    text=True,
                               timeout=timeout,
    encoding="utf-8", errors="replace")
            if not os.path.isfile(pdf_path) or
    os.path.getsize(pdf_path) < 800:
                return None, "浏览器未能生成 PDF（%s）" % (p.stderr or
    "").strip()[:200]
            with open(pdf_path, "rb") as f:
                return f.read(), None
    except subprocess.TimeoutExpired:
            return None, "生成 PDF 超时，请稍后重试"
    except Exception as e:
            return None, "生成 PDF 失败：%s" % e
    finally:
            shutil.rmtree(tmpdir, ignore_errors=True)
# ==== app/services/ahp_service.py ====
# -*- coding: utf-8 -*-
"""AHP 权重引擎。
严格按论文 4.2.1 节实现：
- 式(2)(3)(4)：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式(5)(6)(7)：一致性检验（λmax、CI、CR）
- 论文要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
    """由判断矩阵上三角构造完整互反矩阵。
    参数
    ----
    upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
        例：3 阶时传入 [[a12, a13], [a23]]。
        对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
    """
    upper = np.asarray(upper, dtype=float)
    k = upper.shape[0]
    n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
    A = np.ones((n, n), dtype=float)
    idx = 0
    for i in range(n):
        for j in range(i + 1, n):
            A[i, j] = upper[idx]
```

```
                A[j, i] = 1.0 / upper[idx]
                idx += 1
    return A
def weights_and_consistency(A):
    """由判断矩阵 A 计算权重与一致性指标。
    返回
    ----
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式(2) 行元素乘积
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
    return {
        "weights": weights,
        "lambda_max": lambda_max,
        "ci": ci,
        "cr": cr,
        "n": n,
        "consistent": cr < CR_THRESHOLD,
    }
def find_most_inconsistent(A):
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
    ----
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)
    及对应取值。
    """
    A = np.asarray(A, dtype=float)
    r = weights_and_consistency(A)
    w = r["weights"]
    n = A.shape[0]
    best = None
    for i in range(n):
        for j in range(i + 1, n):
```

```
                expected = w[i] / w[j]
                if expected == 0:
                    continue
                deviation = abs(A[i, j] - expected) / expected
                if best is None or deviation >
    best["deviation"]:
                    best = {"i": i, "j": j, "value": A[i, j],
                            "expected": expected, "deviation":
    deviation}
    return best
def aggregate_matrices(matrices):
    """多专家判断矩阵聚合：几何平均（保持互反性）。
    参数
    ----
    matrices : list[np.ndarray]，多个同阶判断矩阵。
    """
    stack = np.stack([np.asarray(m, dtype=float) for m in
    matrices])
    return np.prod(stack, axis=0) ** (1.0 / len(stack))
def judge_from_weights(weights):
    """由权重向量构造完全一致判断矩阵并检验。
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。
    """
    w = np.asarray(weights, dtype=float)
    n = w.shape[0]
    A = np.outer(w, 1.0 / w)
    r = weights_and_consistency(A)
    r["matrix"] = A
    return r
# ==== app/services/standardization.py ====
# -*- coding: utf-8 -*-
"""指标标准化（的一部分）。
按论文 4.2.2 节式(8) 实现：
    x' = (x - x_min) / (x_max - x_min)
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。
所有理想极值取自 app.utils.constants，可配置。
"""
from app.utils.constants import (
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,
    PH_TOL_LOW, PH_TOL_HIGH,
)
def std_benefit(value, ideal):
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, value / ideal))
```

```
def std_cost(value, extreme):
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, 1.0 - value / extreme))
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,
               tol_hi=PH_TOL_HIGH):
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    论文 4.2.2 节：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（论文 4.2.2 节）。"""
    if aspect is None:
        return None
    return ASPECT_SCORES.get(aspect)
def pollution_score(heavy_metals):
    """污染指标：8 种重金属等权（各 0.125）合成。
    论文注：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染(=1)处理。
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。
    """
    total = 0.0
    for metal, extreme in HEAVY_METAL_LIMIT.items():
        value = heavy_metals.get(metal)
        total += std_cost(value, extreme) if value is not
    None else 1.0
    return total / len(HEAVY_METAL_LIMIT)
def standardize_unit(unit_type, data, weights=None):
    """对单个微地质单元的全部指标做标准化。
    参数
    ----
    unit_type : str，岩质边坡 / 碎石边坡 / 平台
    data : dict，实测数据，字段见接口契约：
        soil_ratio, thickness, ph, organic_matter,
    heavy_metals(dict),
        coverage, shannon, slope, aspect
```

```
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIG
HTS 结构），
                未提供时用默认值。
    返回
    ----
    dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }
    标准化值可能为 None，表示该指标不适用或方法待定。
    """
    from app.utils.constants import DEFAULT_WEIGHTS
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    l1_enabled = weights["l1"].keys()
    l2w = weights["l2"]
    result = {}
    # ---- 土壤指标 ----
    if "土壤指标" in l1_enabled:
        result["土壤指标"] = {
            "土石比": (data.get("soil_ratio"),
                       std_benefit(data.get("soil_ratio"),
IDEAL_SOIL_RATIO),
                       l2w["土壤指标"]["土石比"]),
            "土层厚度": (data.get("thickness"),
                         std_benefit(data.get("thickness"),
IDEAL_THICKNESS),
                         l2w["土壤指标"]["土层厚度"]),
            "pH": (data.get("ph"),
                   std_mid_ph(data.get("ph"),
*IDEAL_PH_RANGE),
                   l2w["土壤指标"]["pH"]),
            "有机质": (data.get("organic_matter"),
                       std_benefit(data.get("organic_matter")
, IDEAL_ORGANIC),
                       l2w["土壤指标"]["有机质"]),
            "污染指标": (data.get("heavy_metals") or {},
                         pollution_score(data.get("heavy_meta
ls") or {}),
                         l2w["土壤指标"]["污染指标"]),
        }
    # ---- 植被指标 ----
    if "植被指标" in l1_enabled:
        result["植被指标"] = {
            "植被覆盖度": (data.get("coverage"),
                           std_benefit(data.get("coverage"),
IDEAL_COVERAGE),
                           l2w["植被指标"]["植被覆盖度"]),
            "生物多样性": (data.get("shannon"),
                           std_benefit(data.get("shannon"),
IDEAL_SHANNON),
                           l2w["植被指标"]["生物多样性"]),
        }
```

```
    #  ----  地形指标  ----                                                    1401
    if  "地形指标"  in  l1_enabled:                                           1402
        result["地形指标"]  =  {                                              1403
            "坡度":  (data.get("slope"),                                      1404
                      std_slope(data.get("slope")),                           1405
                      l2w["地形指标"]["坡度"]),                               1406
            "坡向":  (data.get("aspect"),                                     1407
                      aspect_score(data.get("aspect")),                       1408
                      l2w["地形指标"]["坡向"]),                               1409
        }                                                                     1410
    return  result                                                            1411
#  ====  app/services/weight_service.py  ====                                1412
#  -*-  coding:  utf-8  -*-                                                   1413
"""权重方案的存取与生效权重解析。                                             1414
系统有两套权重来源：                                                         1415
1．  论文表  3  给出的默认权重（app.utils.constants.DEFAULT_WEIGHTS），      1416
    是**权威取值来源**，案例复算与报告结论以此为准；                         1417
2．  通过层次分析法在本系统中实际算出的权重，保存在                          1418
    weight_schemes  /  weight_items  两张表中，可标记为「生效」。            1419
本模块负责把这两者合成为计算引擎直接可用的嵌套权重结构：                    1420
以论文表  3  为底，用生效方案中已保存的条目覆盖。                           1421
没有任何生效方案时，即完全使用论文表  3  的权重。                           1422
这样既保证默认行为与论文一致，又使  AHP  计算结果真正参与评价，             1423
而不是算完即弃。                                                             1424
"""                                                                          1425
import  copy                                                                 1426
from  app.models  import  db,  WeightScheme,  WeightItem                    1427
from  app.utils.constants  import  DEFAULT_WEIGHTS                          1428
from  app.services.indicator_service  import  TREE                          1429
#  方案名称：每个微地质单元类型固定一个可生效的方案                          1430
ACTIVE_NAME  =  "自定义权重方案（AHP  计算结果）"                            1431
def  _code_maps():                                                           1432
    """由指标树导出  编码  →  名称  的映射，避免名称在两处各写一遍。"""      1433
    l1,  l2  =  {},  {}                                                      1434
    for  row  in  TREE:                                                      1435
        code,  name,  level,  parent  =  row[0],  row[1],  row[2],           1436
    row[3]                                                                   1437
        if  level  ==  1:                                                    1438
            l1[code]  =  name                                                1439
        elif  level  ==  2:                                                  1440
            l2[code]  =  (l1.get(parent,  parent),  name)                    1441
    return  l1,  l2                                                          1442
def  effective_weights(unit_type):                                           1443
    """返回该单元类型当前生效的权重。                                        1444
    返回  (weights,  source_label,  scheme)                                  1445
        weights        嵌套权重结构，可直接交给标准化与评价引擎              1446
        source_label   权重来源说明，用于界面与报告标注                     1447
        scheme         生效的权重方案对象，无自定义方案时为  None            1448
    """                                                                      1449
    base  =  DEFAULT_WEIGHTS.get(unit_type)                                  1450
```

```
    if base is None:
        return None, "未定义该微地质单元类型的权重", None
    scheme = WeightScheme.query.filter_by(unit_type=unit_type,
                                          name=ACTIVE_NAME).first()
    if scheme is None:
        return copy.deepcopy(base), "论文表 3 默认权重", None
    merged = copy.deepcopy(base)
    l1_map, l2_map = _code_maps()
    applied = 0
    for item in WeightItem.query.filter_by(scheme_id=scheme.id).all():
        code, w = item.indicator_code, item.weight
        if w is None or w <= 0:
            continue
        if code in l1_map and l1_map[code] in merged["l1"]:
            merged["l1"][l1_map[code]] = w
            applied += 1
        elif code in l2_map:
            l1_name, l2_name = l2_map[code]
            if l1_name in merged["l2"] and l2_name in
    merged["l2"][l1_name]:
                merged["l2"][l1_name][l2_name] = w
                applied += 1
    if applied == 0:
        return copy.deepcopy(base), "论文表 3 默认权重", None
    return merged, "自定义方案（%d 项覆盖论文表 3）" % applied, scheme

def save_weights(unit_type, code_weight_pairs, note=None):
    """把一组「指标编码 → 权重」写入生效方案（已存在则更新对应条目）。
    仅覆盖本次传入的条目，其余条目保留，便于分层次（一级 / 二级）
    分次录入而无需一次填完。
    """
    scheme = WeightScheme.query.filter_by(unit_type=unit_type,
                                          name=ACTIVE_NAME).first()
    if scheme is None:
        scheme = WeightScheme(unit_type=unit_type,
                              name=ACTIVE_NAME,
                              level=1, is_default=True)
        db.session.add(scheme)
        db.session.flush()
    if note:
        scheme.level = scheme.level or 1
    if not scheme.is_default:
        scheme.is_default = True
    written = 0
    for code, w in code_weight_pairs:
        if w is None or w <= 0:
```

```
# ----------------------------------------------------------------------
    ------------------
# 论文第 5 节案例数据（表 5，逐字抄录）
# ----------------------------------------------------------------------
    ------------------
# 三类单元面积占比：岩质边坡 １３．８６%、碎石边坡 ６２．９５%、平台 ２３．１９%
# 注：岩质边坡无土壤指标、平台无地形指标，对应"不适用"。
case = [
    {
        "unit_type": "岩质边坡",
        "ratio": 0.1386,
        "data": {
            "coverage": 68.88, "shannon": 0.00,
            "slope": 63.33, "aspect": "阴阳坡",
        },
    },
    {
        "unit_type": "碎石边坡",
        "ratio": 0.6295,
        "data": {
            "soil_ratio": 0.34, "thickness": 10, "ph":
    9.24, "organic_matter": 4.89,
            "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
    12.60,
                             "Cu": 11.70, "Cd": 0.04, "Pb":
    6.50},
            "coverage": 87.92, "shannon": 0.60,
            "slope": 28.75, "aspect": "阳坡",
        },
    },
    {
        "unit_type": "平台",
        "ratio": 0.2319,
        "data": {
            "soil_ratio": 0.56, "thickness": 30, "ph":
    8.72, "organic_matter": 10.92,
            "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni":
    12.33,
                             "Cu": 11.14, "Cd": 0.09, "Pb":
    9.98},
            "coverage": 58.42, "shannon": 0.72,
        },
    },
]
# ----------------------------------------------------------------------
    ------------------
# 逐单元输出标准化与得分
# ----------------------------------------------------------------------
    ------------------
print(LINE)
```

```
print("玉田县案例 · 评价引擎完整计算")
print(LINE)
for u in case:
    r = ev.score_unit(u["unit_type"], u["data"])
    print()
    print("  【%s】  (面积占比 %.2f%%)" % (u["unit_type"],
    u["ratio"] * 100))
    print("  " + "-" * 60)
    for l1, score in r["l1_scores"].items():
        print("    %s 得分 = %.4f" % (l1, score))
    print("    单元评价因子 F' = %.4f" % r["unit_score"])
print()
print(LINE)
print("面积加权合成（式(9)）")
print(LINE)
res = ev.score_mine(case)
total = res["final_score"]
print()
for ur in res["unit_results"]:
    print("  %s  %.2f%% × %.4f = %.4f"
          % (ur["unit_type"], ur["ratio"] * 100,
    ur["unit_score"], ur["weighted"]))
print("  " + "-" * 60)
print("  最终评价因子 F = %.4f" % total)
print()
print("  适宜性等级：%s" % res["grade"])
print("  判定依据  ：%s" % res["grade_info"]["meaning"])
print("  推荐措施  ：%s" % res["grade_info"]["measure"])
print()
print(LINE)
print("结论")
print(LINE)
print("  - 评价引擎按论文式(8)(9) 与表 3 权重完整跑通")
print("  - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")
print("  - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"
      % (total, res["grade"]))
print("  - 论文表 5 所载最终得分为 0.76；差异集中在坡度、坡向的标准化口径，")
print("    该口径在 standardization.std_slope / aspect_score
    中集中定义、可配置")
print(LINE)
#  ==== 测试/test_web_ahp.py ====
#  -*- coding: utf-8 -*-
"""AHP 网页界面自检：走通选择→录入→计算 全流程。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import os
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
```

```
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
# 测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
        db.session.add(User(username="admin",
                            password_hash=generate_password_hash("admin123"),
                            role="admin"))
        db.session.commit()
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  首页与 AHP 入口")
print(LINE)
r = client.get("/")
print("  首页状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
r = client.get("/ahp/")
print("  AHP 入口状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
assert "微地质单元类型" in r.get_data(as_text=True)
print()
print(LINE)
print("测试 2  选择碎石边坡一级指标 → 跳转矩阵页")
print(LINE)
r = client.post("/ahp/", data={
    "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转后页面含「判断矩阵录入」：", "判断矩阵录入" in body)
print("  含三个一级指标（土壤/植被/地形）：", all(n in body for n in
    ("土壤指标", "植被指标", "地形指标")))
assert "判断矩阵录入" in body
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))
print()
```

```
print(LINE)
print("测试 3  提交论文表 3 一致矩阵 → 还原权重")
print(LINE)
# 碎石边坡一级权重 （0.44， 0.31， 0.25） 对应的完全一致矩阵上三角：
# a12 = 0.44/0.31 = 1.4194， a13 = 0.44/0.25 = 1.76， a23 =
    0.31/0.25 = 1.24
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  结果页含「计算结果」:", "计算结果" in body)
print("  含权重 0.44:", "0.44" in body, " 含 0.31:", "0.31" in
    body, " 含 0.25:", "0.25" in body)
print("  含一致性检验通过:", "通过" in body)
assert "计算结果" in body
assert "0.44" in body and "0.31" in body and "0.25" in body
print()
print(LINE)
print("测试 4  不一致矩阵应提示不通过并定位")
print(LINE)
# 明显不一致：a12=9， a13=1， a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  含「不通过」:", "不通过" in body)
print("  含「建议优先调整」:", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
print()
print(LINE)
print("网页界面全部测试通过 √")
print(LINE)
# ==== 测试/test_web_modules.py ====
# -*- coding: utf-8 -*-
"""模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__
    )))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
```

```
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
print(LINE)
print("测试 1  未登录应被重定向到登录页")
print(LINE)
r = client.get("/project/")
print("  /project/ 状态码:", r.status_code, " 重定向到:",
    r.headers.get("Location"))
assert r.status_code == 302 and "login" in
    r.headers.get("Location", "")
print()
print(LINE)
print("测试 2  登录")
print(LINE)
r = client.post("/login", data={"username": "admin",
    "password": "admin123"},
                follow_redirects=True)
body = r.get_data(as_text=True)
# 工作台:页首横幅 + 评价流程分区 + 数据概览
print("  登录后含「评价流程」:", "评价流程" in body)
print("  含账户 admin:", "admin" in body)
print("  含页首横幅:", 'class="hero"' in body)
print("  含数据概览:", 'class="stat-card"' in body)
assert "评价流程" in body and 'class="hero"' in body
print()
print(LINE)
print("测试 3  项目与矿山档案")
print(LINE)
r = client.get("/project/")
body = r.get_data(as_text=True)
print("  项目列表状态码:", r.status_code)
print("  含案例项目:", "玉田县典型历史遗留废弃矿山案例" in body)
assert "玉田县典型历史遗留废弃矿山案例" in body
# 新建项目
client.post("/project/add", data={"name": "测试项目A", "code":
    "T-001"})
r = client.get("/project/")
print("  新建项目后含「测试项目A」:", "测试项目A" in r.get_data(as_text=True
    ))
assert "测试项目A" in r.get_data(as_text=True)
# 清理测试数据,避免污染演示库
with app.app_context():
    from app.models import db, Project
    for p in Project.query.filter_by(name="测试项目A").all():
        db.session.delete(p)
    db.session.commit()
```

```
print("    已清理测试项目")
print()
print(LINE)
print("测试 4    微地质单元（面积占比自动计算）")
print(LINE)
from app.models import MinePatch
with app.app_context():
    patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").f
    irst()
    pid = patch.id
r = client.get("/unit/%d" % pid)
body = r.get_data(as_text=True)
print("    单元列表含三类单元：", all(t in body for t in ("岩质边坡",
    "碎石边坡", "平台")))
print("    含占比合计：", "占比合计" in body)
assert "岩质边坡" in body and "占比合计" in body
print()
print(LINE)
print("测试 5    标准知识库与省际口径比对")
print(LINE)
r = client.get("/standard/")
body = r.get_data(as_text=True)
print("    含辽宁/湖南/新疆标准：", all(p in body for p in ("辽宁", "湖南",
    "新疆")))
assert "辽宁" in body
r = client.get("/standard/compare")
body = r.get_data(as_text=True)
print("    口径比对页含「植被覆盖度」分组：", "植被覆盖度" in body)
print("    含多省份并列：", "湖南" in body and "新疆" in body)
assert "植被覆盖度" in body
print()
print(LINE)
print("测试 6    限制性指标判定（一票否决）")
print(LINE)
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("    含五项条件：", all(x in body for x in ("政策约束", "区位约束",
    "土壤污染", "地质灾害", "耕地属性")))
assert all(x in body for x in ("政策约束", "区位约束", "土壤污染",
    "地质灾害", "耕地属性"))
# 全部通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("    全通过后显示「通过，可进入定量评价」：", "可进入定量评价" in body)
assert "可进入定量评价" in body
# 一项不通过
```

```
client.post("/constraint/%d" % pid, data={                              11103
    "policy_pass": "1", "location_pass": "0",                           11104
    "pollution_pass": "1",                                              11105
    "geo_pass": "1", "farmland_pass": "1"})                             11106
r = client.get("/constraint/%d" % pid)                                  11107
body = r.get_data(as_text=True)                                         11108
print("  一项不通过后显示「不通过」:", "不通过" in body)                    11109
print("  含失败项「区位约束」:", "区位约束" in body)                        11110
assert "不通过" in body                                                  11111
# 恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤                  11112
client.post("/constraint/%d" % pid, data={                              11113
    "policy_pass": "1", "location_pass": "1",                           11114
    "pollution_pass": "1",                                              11115
    "geo_pass": "1", "farmland_pass": "1"})                             11116
r = client.get("/constraint/%d" % pid)                                  11117
print("  已恢复为全部通过:", "可进入定量评价" in r.get_data(as_text=True))  11118
print()                                                                 11119
print(LINE)                                                             11120
print("全部测试通过 √")                                                  11121
print(LINE)                                                             11122
# ==== 测试/test_web_evaluate.py ====                                   11123
# -*- coding: utf-8 -*-                                                 11124
"""端到端自检：录数据 → 执行评价 → 查看结果。"""                            11125
import os                                                               11126
import sys                                                              11127
sys.stdout.reconfigure(encoding="utf-8")                                11128
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，                    11129
# 使 `from app import ...` 与 app/app.db 的定位不受影响。                 11130
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil            11131
    e__)))                                                              11132
if ROOT not in sys.path:                                                11133
    sys.path.insert(0, ROOT)                                            11134
from app import create_app                                              11135
LINE = "=" * 66                                                         11136
# 与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端                    11137
class TestConfig:                                                       11138
    SECRET_KEY = "test"                                                 11139
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(              11140
        ROOT, "app", "app.db")                                          11141
    SQLALCHEMY_TRACK_MODIFICATIONS = False                              11142
    WTF_CSRF_ENABLED = False                                            11143
    TESTING = True                                                      11144
app = create_app(TestConfig)                                            11145
client = app.test_client()                                              11146
# 登录（案例种子已创建 admin 用户）                                        11147
client.post("/login", data={"username": "admin",                        11148
    "password": "admin123"})                                            11149
print(LINE)                                                             11150
print("测试 1  指标数据列表")                                            11151
print(LINE)                                                             11152
```

```
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含案例图斑：", "玉田县孤树镇白云岩采石场" in body)
print("  含三个单元：", all(n in body for n in ("岩质边坡", "碎石边坡",
    "平台")))
assert r.status_code == 200
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body
print()
print(LINE)
print("测试 2  指标数据录入页（按单元类型显示字段）")
print(LINE)
from app.models import MicroUnit, MinePatch
with app.app_context():
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡").
    first()
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡"
    ).first()
r = client.get("/indicator/%d" % unit_rock.id)
b_rock = r.get_data(as_text=True)
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）")
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock)
assert "土石比" not in b_rock and "植被覆盖度" in b_rock
r = client.get("/indicator/%d" % unit_gravel.id)
b_gravel = r.get_data(as_text=True)
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel)
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")))
assert "土石比" in b_gravel and all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))
print()
print(LINE)
print("测试 3  执行评价")
print(LINE)
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
r = client.post("/evaluate/%d" % patch.id, data={},
    follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转至结果页：", "适宜性评价结果" in body)
print("  含最终评价因子 F：", "最终评价因子 F" in body)
print("  含等级「适宜」：", "适宜" in body)
print("  含推荐措施「自然恢复」：", "自然恢复" in body)
assert "最终评价因子 F" in body and "自然恢复" in body
print()
print(LINE)
print("测试 4  结果持久化")
print(LINE)
from app.models import EvaluationResult
```

```
with app.app_context():
    res = EvaluationResult.query.order_by(EvaluationResult.i
    d.desc()).first()
    print("  最终得分 F =", res.final_score)
    print("  等级 =", res.grade)
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("端到端全部测试通过 √")
print(LINE)
#  ==== 测试/test_web_report.py ====
#  -*- coding: utf-8 -*-
"""报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil
    e__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
#  登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
from app.models import MinePatch, EvaluationResult
with app.app_context():
    import seed_demo
    patch = seed_demo.demo_patch()
    pid = patch.id
#  报告取自已保存的评价记录，因此先按网页路径真实执行一次评价
from app.views.evaluate import load_patch_input,
    save_evaluation
with app.app_context():
    units = load_patch_input(pid)
    rec = save_evaluation(pid, units)
    rid = rec.id
```

```
        print("前置：已执行评价并保存记录 id=%d, F=%.4f (%s)"                    11253
              % (rid, rec.final_score, rec.grade))                              11254
        from app.models import ScoreDetail                                      11255
        n_detail = ScoreDetail.query.filter_by(result_id=rid).co                11256
        unt()                                                                   11257
        print("          逐层得分明细已落库：%d 条" % n_detail)                  11258
print()                                                                         11259
print(LINE)                                                                     11260
print("测试 1  报告列表")                                                      11261
print(LINE)                                                                     11262
r = client.get("/report/")                                                      11263
print("  状态码：", r.status_code)                                             11264
body = r.get_data(as_text=True)                                                 11265
print("  含「所属项目」列：", "所属项目" in body)                              11266
assert r.status_code == 200                                                     11267
print()                                                                         11268
print(LINE)                                                                     11269
print("测试 2  得分明细页")                                                    11270
print(LINE)                                                                     11271
r = client.get("/report/%d" % pid)                                              11272
body = r.get_data(as_text=True)                                                 11273
print("  状态码：", r.status_code)                                             11274
print("  含「得分明细」：", "得分明细" in body)                                11275
print("  含「最终评价因子 F」：", "最终评价因子 F" in body)                    11276
print("  含二级指标明细（植被覆盖度/坡度/坡向）：", all(n in body for n in     11277
      ("植被覆盖度", "坡度", "坡向")))                                          11278
print("  含权重/标准化值列：", "标准化值" in body and "贡献" in body)          11279
assert r.status_code == 200                                                     11280
assert "植被覆盖度" in body and "标准化值" in body                              11281
print()                                                                         11282
print(LINE)                                                                     11283
print("测试 3  Word 报告导出")                                                 11284
print(LINE)                                                                     11285
r = client.get("/report/%d/word" % pid)                                         11286
print("  状态码：", r.status_code)                                             11287
print("  Content-Type:", r.headers.get("Content-Type",                          11288
      "")[:60])                                                                 11289
data = r.data                                                                   11290
print("  文件大小：", len(data), "字节")                                       11291
print("  是否为有效 docx（PK 头）：", data[:2] == b"PK")                       11292
assert r.status_code == 200 and data[:2] == b"PK"                               11293
# 用 python-docx 打开验证内容                                                  11294
import docx                                                                     11295
d = docx.Document(io.BytesIO(data))                                             11296
text = "\n".join(p.text for p in d.paragraphs)                                  11297
print("  含报告标题：", "自然恢复适宜性评价报告" in text)                      11298
print("  含评价结论章节：", "评价结论" in text)                                11299
print("  含得分明细章节：", "得分明细" in text)                                11300
print("  段落数：", len(d.paragraphs), " 表格数：", len(d.tables))             11301
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3                  11302
```

```
print()
print(LINE)
print("报告输出全部测试通过 √")
print(LINE)
# ==== 测试/test_ui_fixes.py ====
# -*- coding: utf-8 -*-
"""界面修复自检：「记住我」与各模块删除功能。"""
import os
import re
import sys
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
sys.stdout.reconfigure(encoding="utf-8")
LINE = "=" * 70
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        ROOT, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    REMEMBER_COOKIE_DURATION = __import__("datetime").timedelta(days=30)
app = create_app(TestConfig)
client = app.test_client()
# ------------------------------------------------------------------
print(LINE)
print("一、「记住我」")
print(LINE)
html = client.get("/login").get_data(as_text=True)
# 1) 复选框必须有可点击关联的 label（这是原先失效的根因）
#    只判断「复选框与文字是否落在同一个 <label> 内」，不绑定具体类名 ——
#    否则换个样式类就会误报，而真正的回归点其实与类名无关。
labels = re.findall(r"<label[^>]*>(.*?)</label>", html, re.S)
wrapped = any(('name="remember"' in b and "记住我" in b) for b in labels)
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped else "[不符合]"))
assert wrapped, "复选框未与文字关联，点击文字不会勾选"
has_for = 'for="remember"' in html
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else "[不符合]"))
# 2) 勾选后应下发带过期时间的 remember_token
```

```
c = app.test_client()
r = c.post("/login", data={"username": "admin", "password":
    "admin123",
                               "remember": "y"},
    follow_redirects=False)
cookies = r.headers.getlist("Set-Cookie")
rt = [h for h in cookies if h.startswith("remember_token=")]
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else
    "[不符合]"))
assert rt, "未下发 remember_token"
exp = re.search(r"Expires=([^;]+)", rt[0])
print("  %s 该 cookie 带过期时间：%s"
      % ("[符合]" if exp else "[不符合]", exp.group(1) if exp
    else "无（会话 cookie）"))
assert exp, "remember_token 无过期时间，关浏览器即失效"
# 3) 不勾选时不应下发 remember_token
c2 = app.test_client()
r2 = c2.post("/login", data={"username": "admin",
    "password": "admin123"},
                follow_redirects=False)
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if
    h.startswith("remember_token=")]
print("  %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2
    else "[不符合]"))
assert not rt2
# 4) 携带 remember_token 的新会话应自动登录
c3 = app.test_client()
c3.set_cookie("remember_token",
              rt[0].split(";")[0].split("=", 1)[1],
    domain="localhost")
r3 = c3.get("/", follow_redirects=False)
print("  %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"
      % ("[符合]" if r3.status_code == 200 else "[不符合]",
    r3.status_code))
assert r3.status_code == 200
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("二、删除功能")
print(LINE)
client.post("/login", data={"username": "admin",
    "password": "admin123"})
# 项目：列表页须有删除按钮
html = client.get("/project/").get_data(as_text=True)
has_del = "/delete" in html and "btn--danger" in html
print("  %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))
assert has_del, "项目列表缺少删除按钮"
print("  %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else
    "[不符合]"))
```

```
assert "confirm(" in html                                                    11403
# 图斑列表：地址用的是项目主键，不是图斑主键                                  11404
with app.app_context():                                                      11405
    from app.models import MinePatch, Project                                11406
    import seed_demo                                                         11407
    patch = seed_demo.demo_patch()                                           11408
    patch_id = patch.id                                                      11409
    project_id = patch.project_id                                            11410
html = client.get("/project/%d/patches" %                                    11411
    project_id).get_data(as_text=True)                                       11412
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in          11413
    html and "btn--danger" in html else "[不符合]"))                         11414
assert "/project/patch/" in html and "btn--danger" in html                   11415
# 微地质单元                                                                 11416
html = client.get("/unit/%d" % patch_id).get_data(as_text=Tr                 11417
    ue)                                                                      11418
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in         11419
    html else "[不符合]"))                                                   11420
assert "/unit/delete/" in html                                               11421
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"                    11422
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))                 11423
# 标准                                                                       11424
html = client.get("/standard/").get_data(as_text=True)                       11425
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html        11426
    else "[不符合]"))                                                        11427
assert "btn--danger" in html                                                 11428
# 实际执行一次删除：新建项目 → 删除 → 确认消失                               11429
client.post("/project/add", data={"name": "待删除测试项目",                 11430
    "code": "DEL-1"})                                                        11431
with app.app_context():                                                      11432
    from app.models import db, Project                                       11433
    tmp = Project.query.filter_by(name="待删除测试项目").first()             11434
    tmp_id = tmp.id                                                          11435
html = client.get("/project/").get_data(as_text=True)                        11436
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else       11437
    "[不符合]"))                                                             11438
r = client.post("/project/%d/delete" % tmp_id,                               11439
    follow_redirects=True)                                                   11440
html = r.get_data(as_text=True)                                              11441
gone = "待删除测试项目" not in html                                          11442
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))  11443
assert gone                                                                  11444
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else            11445
    "[不符合]"))                                                             11446
print()                                                                      11447
print(LINE)                                                                  11448
print("界面修复自检全部通过 √")                                             11449
print(LINE)                                                                  11450
# ==== 测试/test_new_features.py ====                                        11451
# -*- coding: utf-8 -*-                                                      11452
```

```
"""补齐模块自检：操作日志与用户维护、遥感影像、指标与多专家矩阵、
批量导入、图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
import os
import re
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
# 本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
# 使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                        Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    # 固定取内置演示案例的图斑，避免受库中用户自建项目 / 验证算例的影响
    import seed_demo
    patch_id = seed_demo.demo_patch().id
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
```

```
def ok(flag, label, extra=""):
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    assert flag, label
# ------------------------------------------------------------------
    ------------------
print(LINE)
print("操作日志与用户维护")
print(LINE)
before = None
with app.app_context():
    before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code":
    "LOG-T"})
with app.app_context():
    after = OperationLog.query.count()
    last = OperationLog.query.order_by(OperationLog.id.desc(
    )).first()
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,
    after))
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
ok("project" in (last.target_type or ""), "日志记录了对象类型",
    last.target_type)
r = c.get("/profile?tab=logs")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "操作日志" in body, "操作日志分类可访问")
ok("新建项目" in body, "日志页显示操作记录")
# 账号编辑 / 删除（职务只有管理员与操作员两种）
with app.app_context():
    u = User(username="tmp_user", role="operator",
             password_hash=User.query.first().password_hash)
    db.session.add(u); db.session.commit()
    uid = u.id
r = c.get("/users/%d/edit" % uid)
ok(r.status_code == 200, "账号编辑页可访问")
# 保护校验要趁系统里只有一名管理员时做：
# 先降级唯一的 admin 应被拒绝，再把临时账号提为管理员看修改是否生效
with app.app_context():
    admin_id = User.query.filter_by(role="admin").first().id
c.post("/users/%d/edit" % admin_id,
       data={"role": "operator", "active": "1"})
with app.app_context():
    ok(db.session.get(User, admin_id).role == "admin",
    "拒绝把最后一个管理员降级")
c.post("/users/%d/edit" % uid, data={"role": "admin",
    "active": "1"})
with app.app_context():
    ok(db.session.get(User, uid).role == "admin",
    "职务已修改并保存")
c.post("/users/%d/delete" % uid, follow_redirects=True)
```

```
with app.app_context():
    ok(db.session.get(User, uid) is None, "账号删除生效")
# 清理日志核对用的项目
with app.app_context():
    for p in Project.query.filter_by(name="日志核对项目").all():
        db.session.delete(p)
    db.session.commit()
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("遥感影像挂接")
print(LINE)
doc = fitz.open()
pg = doc.new_page(width=320, height=200)
pg.insert_text((20, 60), "RS image", fontname="china-s",
    fontsize=16)
png = doc[0].get_pixmap(dpi=100).tobytes("png")
doc.close()
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(png), "影像.png"),
                 "image_note": "2025年6月无人机正射影像"},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
    rel = db.session.get(MinePatch, patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
    rel.replace("uploads/", "", 1))
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(b"x"), "bad.txt")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
    follow_redirects=True)
with app.app_context():
    ok(db.session.get(MinePatch, patch_id).image_path is
    None, "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("指标定义维护与多专家矩阵")
```

```
print(LINE)
r = c.get("/ahp/indicators")
body = r.get_data(as_text=True)
with app.app_context():
    n_ind = Indicator.query.count()
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d
    项" % n_ind)
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),
   "指标树按层级展示")
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")
c.post("/ahp/indicators/coverage/edit",
       data={"name": "植被覆盖度", "unit": "%", "ideal_type":
    "benefit",
             "ideal_value": "75", "enabled_units": ["岩质边坡"],
             "description": "自检修改"})
with app.app_context():
    ok(Indicator.query.filter_by(code="coverage").first().de
    scription == "自检修改",
       "指标定义可编辑")
c.post("/ahp/indicators/reset")
with app.app_context():
    ok("自检修改" not in (Indicator.query.filter_by(code="covera
    ge").first().description or ""),
       "可按论文取值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
    d = dict(base); d.update({"action": "add", "unit_type":
    "碎石边坡",
                              "level": "1", "expert_name":
    name})
    c.post("/ahp/experts", data=d)
with app.app_context():
    ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡",
    level=1).count() == 3,
       "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
   "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>",
    body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws
    else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) <
    0.01,
   "汇总权重归一", "、".join(vals))
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
    vals[0] if vals else "")
# ------------------------------------------------------------------
    ----------------
```

```
print()
print(LINE)
print("批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
    "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度
    ,坡向\n"
    "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,
    87.92,0.60,28.75,阳坡\n"
    "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,5
    8.42,0.72,,,\n"
    "不存在单元,1,1,7,1,,,,,,,,,\n"
)
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(csv_text.encode("ut
    f-8")), "d.csv")},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("导入结果" in body, "导入结果页展示")
ok("不存在单元" in body, "错误行被报告")
with app.app_context():
    u = MicroUnit.query.filter_by(patch_id=patch_id,
    unit_type="碎石边坡").first()
    ind = IndicatorData.query.filter_by(unit_id=u.id).first(
    )
    ok(abs(ind.soil_ratio - 0.34) < 1e-6 and
    abs(ind.coverage - 87.92) < 1e-6,
        "导入数据已写入数据库")
r = c.post("/indicator/import",
           data={"patch_id": str(patch_id),
                 "datafile": (io.BytesIO(b"x"), "d.pdf")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")
# ----------------------------------------------------------------
    ------------------
print()
print(LINE)
print("图表与 PDF 导出")
print(LINE)
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,
    "0.8376")])
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,
```

```
        "0.4000")])                                                    11703
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")                    11704
#  报告是评价结论的呈现，需先真实执行一次评价才能打开                          11705
with app.app_context():                                                11706
    from app.views.evaluate import load_patch_input,                   11707
    save_evaluation                                                    11708
    save_evaluation(patch_id, load_patch_input(patch_id))              11709
r = c.get("/report/%d" % patch_id)                                     11710
body = r.get_data(as_text=True)                                        11711
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %           11712
    body.count("<svg"))                                                11713
exe = browser.browser_path()                                           11714
if exe is None:                                                        11715
    print("  [跳过] 未找到浏览器，PDF 导出无法验证")                       11716
else:                                                                  11717
    r = c.get("/report/%d/pdf" % patch_id)                             11718
    data = r.data                                                      11719
    ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %            11720
    len(data))                                                         11721
    d = fitz.open(stream=data, filetype="pdf")                         11722
    txt = "".join(d[i].get_text() for i in                             11723
    range(d.page_count))                                               11724
    ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")                  11725
    ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")     11726
    ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)      11727
    d.close()                                                          11728
print()                                                                11729
print(LINE)                                                            11730
print("补齐模块自检全部通过 √")                                          11731
print(LINE)                                                            11732
#  ==== 测试/test_workflow.py ====                                     11733
#  -*- coding: utf-8 -*-                                               11734
"""流程门禁自检：前置步骤未完成时，后续模块必须拦得住。                      11735
论文 4.1 节确定的评价流程存在先后关系，缺任何一环都不应产出评价结论。          11736
本测试逐个构造「缺前置」的场景，验证系统不会放行。                          11737
"""                                                                    11738
import os                                                              11739
import sys                                                             11740
sys.stdout.reconfigure(encoding="utf-8")                               11741
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，                 11742
#  使 `from app import ...` 与 app/app.db 的定位不受影响。               11743
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__fil           11744
    e__)))                                                             11745
if ROOT not in sys.path:                                               11746
    sys.path.insert(0, ROOT)                                           11747
from app import create_app                                             11748
from app.models import (db, Project, MinePatch, MicroUnit,             11749
    IndicatorData,                                                     11750
                            ConstraintCheck, EvaluationResult)         11751
from app.services import workflow as wf                                11752
```

```
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
def new_patch(tag):
    with app.app_context():
        p = Project(name="门禁-" + tag, code="WF-" + tag)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="门禁图斑-" +
    tag)
        db.session.add(patch); db.session.flush()
        pid = patch.id
        db.session.commit()
    return pid
def add_units(pid):
    with app.app_context():
        db.session.add(MicroUnit(patch_id=pid,
    unit_type="碎石边坡",
                                 area=1.0, ratio=1.0))
        db.session.commit()
def add_constraint(pid, passed=True):
    with app.app_context():
        db.session.add(ConstraintCheck(
            patch_id=pid, policy_pass=True,
    location_pass=passed,
            pollution_pass=True, geo_pass=True,
    farmland_pass=True,
            passed=passed, reason="全部通过" if passed else
    "未通过：区位约束"))
        db.session.commit()
def add_data(pid):
    with app.app_context():
        u = MicroUnit.query.filter_by(patch_id=pid).first()
        db.session.add(IndicatorData(unit_id=u.id,
    soil_ratio=0.34, thickness=10,
```

```
                                                ph=9.24,                    11803
        organic_matter=4.89, coverage=87.92,                                11804
                                                shannon=0.60,               11805
        slope=28.75, aspect="阳坡"))                                        11806
            db.session.commit()                                             11807
def cleanup(pid, tag):                                                      11808
        with app.app_context():                                             11809
            from app.models import ScoreDetail                              11810
            # 先删得分明细，再删评价记录，避免留下孤儿明细                  11811
            rids = [r.id for r in EvaluationResult.query.filter_            11812
        by(patch_id=pid)]                                                   11813
            if rids:                                                        11814
                ScoreDetail.query.filter(ScoreDetail.result_id.i            11815
        n_(rids)).delete(                                                   11816
                    synchronize_session=False)                              11817
            EvaluationResult.query.filter_by(patch_id=pid).delet            11818
        e()                                                                 11819
            ConstraintCheck.query.filter_by(patch_id=pid).delete            11820
        ()                                                                  11821
            uids = [u.id for u in MicroUnit.query.filter_by(pat             11822
        h_id=pid)]                                                          11823
            if uids:                                                        11824
                IndicatorData.query.filter(IndicatorData.unit_id            11825
        .in_(uids)).delete(                                                 11826
                    synchronize_session=False)                              11827
            MicroUnit.query.filter_by(patch_id=pid).delete()                11828
            m = db.session.get(MinePatch, pid)                              11829
            if m:                                                           11830
                db.session.delete(m)                                        11831
            pr = Project.query.filter_by(name="门禁-" +                     11832
        tag).first()                                                        11833
            if pr:                                                          11834
                db.session.delete(pr)                                       11835
            db.session.commit()                                             11836
def results_of(pid):                                                        11837
        with app.app_context():                                             11838
            return EvaluationResult.query.filter_by(patch_id=pid            11839
        ).count()                                                           11840
def post_eval(pid):                                                         11841
        return c.post("/evaluate/%d" % pid, data={},                        11842
        follow_redirects=False)                                             11843
def get_report(pid):                                                        11844
        return c.get("/report/%d" % pid, follow_redirects=False)            11845
# ------------------------------------------------------------------------  11846
        ----------------                                                    11847
print(LINE)                                                                 11848
print("一、前置条件缺失时，评价必须被拦住")                                 11849
print(LINE)                                                                 11850
print()                                                                     11851
print("场景 1：连微地质单元都没划分")                                       11852
```

```
pid = new_patch("1")
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未划分单元 → 不产生评价结果",
   "HTTP %d，结果数 %d" % (r.status_code, n))
with app.app_context():
    pr = wf.patch_progress(pid)
steps = [b["step"] for b in pr["blocking"]]
ok(not pr["ready"] and "units" in steps,
   "拦原因包含「划分微地质单元」", "、".join(steps))
ok(steps == ["constraint", "units"],
   "拦截顺序符合论文 4.1 节：限制性判定在前，划分单元在后",
   "、".join(steps))
cleanup(pid, "1")
print()
print("场景 2：有单元，但未做限制性条件判定")
pid = new_patch("2"); add_units(pid); add_data(pid)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "未做限制性判定 → 不产生评价结果",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(any(b["step"] == "constraint" for b in pr["blocking"]),
   "拦原因指向「执行限制性条件判定」")
cleanup(pid, "2")
print()
print("场景 3：限制性条件判定【未通过】")
pid = new_patch("3"); add_units(pid); add_data(pid);
    add_constraint(pid, passed=False)
r = post_eval(pid)
n = results_of(pid)
ok(r.status_code == 302 and n == 0, "判定未通过 → 不进入定量评价",
   "结果数 %d" % n)
with app.app_context():
    pr = wf.patch_progress(pid)
ok(not pr["constraint"]["passed"], "流程状态显示判定未通过")
blocked = [b for b in pr["blocking"] if b["step"] ==
    "constraint"]
# 提示只写「缺什么」：label 说明未通过，reason 直接引用判定记录里没过的项
ok(bool(blocked) and "未通过" in blocked[0]["label"],
   "拦提示指明限制性条件判定未通过", blocked[0]["label"] if blocked else
    "")
ok(bool(blocked) and "区位约束" in (blocked[0]["reason"] or ""),
   "并把具体没通过的条件带出来", blocked[0]["reason"] if blocked else "")
cleanup(pid, "3")
print()
print("场景 4：判定通过，但指标数据全空")
pid = new_patch("4"); add_units(pid); add_constraint(pid,
    passed=True)
```

```
r = post_eval(pid)                                                    11903
n = results_of(pid)                                                   11904
ok(r.status_code == 302 and n == 0, "数据未录入 → 不产生评价结果",      11905
   "结果数 %d" % n)                                                   11906
with app.app_context():                                               11907
    pr = wf.patch_progress(pid)                                       11908
ok(any(b["step"] == "data" for b in pr["blocking"]),                  11909
   "拦原因指向「录入指标数据」")                                      11910
ok(pr["data"]["missing"] == ["碎石边坡"],                             11911
   "指出具体缺哪个单元的数据", "、".join(pr["data"]["missing"]))      11912
cleanup(pid, "4")                                                     11913
print()                                                               11914
print("场景 5：单元面积占比合计不为 100%")                           11915
pid = new_patch("5"); add_constraint(pid, passed=True)                11916
with app.app_context():                                               11917
    u = MicroUnit(patch_id=pid, unit_type="碎石边坡", area=1.0,       11918
    ratio=0.5)                                                        11919
    db.session.add(u); db.session.commit()                            11920
    db.session.add(IndicatorData(unit_id=u.id,                        11921
    soil_ratio=0.3, thickness=10,                                     11922
                                          ph=8.0,                     11923
    organic_matter=5.0, coverage=80.0,                                11924
                                          shannon=0.6, slope=20.0,    11925
    aspect="阳坡"))                                                   11926
    db.session.commit()                                               11927
r = post_eval(pid)                                                    11928
ok(results_of(pid) == 0, "占比不足 100% → 不产生评价结果")            11929
cleanup(pid, "5")                                                     11930
print()                                                               11931
print(LINE)                                                           11932
print("二、前置条件齐备时才放行")                                     11933
print(LINE)                                                           11934
pid = new_patch("OK"); add_units(pid); add_constraint(pid,            11935
    passed=True); add_data(pid)                                       11936
with app.app_context():                                               11937
    pr = wf.patch_progress(pid)                                       11938
ok(pr["ready"], "单元 + 判定通过 + 数据齐备 → 判定为可执行")          11939
r = post_eval(pid)                                                    11940
n = results_of(pid)                                                   11941
ok(n == 1, "此时执行评价成功产生结果", "结果数 %d" % n)               11942
print("  重定向到：", r.headers.get("Location"))                      11943
print()                                                               11944
print(LINE)                                                           11945
print("三、报告不得先于评价产生")                                     11946
print(LINE)                                                           11947
pid2 = new_patch("R")    # 什么都不做                                 11948
r = get_report(pid2)                                                  11949
ok(r.status_code == 302 and "/report/" ==                             11950
    (r.headers.get("Location") or ""),                                11951
   "前置缺失时 /report/<id> 被退回列表", "HTTP %d" % r.status_code)   11952
```

```
for suffix in ("/pdf", "/word"):                                    11953
    rr = c.get("/report/%d%s" % (pid2, suffix),                     11954
    follow_redirects=False)                                         11955
    ok(rr.status_code == 302, "导出 %s 同样被拦住" % suffix, "HTTP  11956
    %d" % rr.status_code)                                           11957
cleanup(pid2, "R")                                                  11958
cleanup(pid, "OK")                                                  11959
print()                                                             11960
print(LINE)                                                         11961
print("四、前置齐备但从未执行评价时，报告仍不得产生")                  11962
print(LINE)                                                         11963
#  报告是评价结论的呈现：即使前置步骤都做完了，只要没真正执行过定量评价，  11964
#  也不应凭空推算出结论。                                           11965
pid3 = new_patch("N")                                               11966
add_units(pid3); add_constraint(pid3, passed=True);                 11967
    add_data(pid3)                                                  11968
with app.app_context():                                             11969
    pr = wf.report_progress(pid3)                                   11970
ok(not pr["ready"], "无评价记录 → 报告流程判定为未就绪")            11971
ok(bool(pr["blocking"]) and pr["blocking"][0]["step"] ==            11972
    "evaluate",                                                     11973
   "拦原因指向「执行适宜性评价」")                                  11974
r = get_report(pid3)                                                11975
ok(r.status_code == 302 and results_of(pid3) == 0,                  11976
   "直接访问 /report/<id> 被退回，且不产生任何评价结果",            11977
   "HTTP %d" % r.status_code)                                       11978
#  真正执行评价之后，报告才放行，且取值与评价记录一致              11979
r = post_eval(pid3)                                                 11980
ok(results_of(pid3) == 1, "先执行评价 → 产生评价记录")              11981
with app.app_context():                                             11982
    pr = wf.report_progress(pid3)                                   11983
    rec = pr["result"]                                              11984
ok(pr["ready"] and rec is not None, "有评价记录后，报告流程就绪")   11985
rr = get_report(pid3)                                               11986
body = rr.get_data(as_text=True)                                    11987
ok(rr.status_code == 200, "此时可打开得分明细页", "HTTP %d" %       11988
    rr.status_code)                                                 11989
ok(("%.4f" % rec.final_score) in body,                              11990
   "报告数值取自评价记录，而非重新计算",                            11991
   "F=%.4f" % rec.final_score)                                      11992
cleanup(pid3, "N")                                                  11993
print()                                                             11994
print(LINE)                                                         11995
print("五、内置演示数据的清理不得波及用户自建数据")                  11996
print(LINE)                                                         11997
#  回归保护：seed_demo.clean_case() 曾执行 Project.query.delete(),  11998
#  导致用户新建的项目在每次启动时被连带删除。                      11999
import seed_demo as sd                                              12000
with app.app_context():                                             12001
    keep_p = Project(name="用户自建项目-守护", code="KEEP-001")      12002
```

```
    db.session.add(keep_p); db.session.flush()
    keep_patch = MinePatch(project_id=keep_p.id,
    name="用户自建图斑 - 守护")
    db.session.add(keep_patch); db.session.flush()
    keep_pid = keep_p.id
    keep_patch_id = keep_patch.id
    db.session.commit()
with app.app_context():
    sd.clean_case(reset=False)
with app.app_context():
    still_p = db.session.get(Project, keep_pid)
    still_patch = db.session.get(MinePatch, keep_patch_id)
ok(still_p is not None, "清理演示数据后，用户自建项目仍存在")
ok(still_patch is not None, "清理演示数据后，用户自建图斑仍存在")
# 演示数据清掉后应能重新写入，保证演示环境可恢复
sd.seed()
with app.app_context():
    demo_back = Project.query.filter_by(code=sd.DEMO_CODE).f
    irst()
ok(demo_back is not None, "演示案例可重新写入（幂等恢复）")
with app.app_context():
    m = db.session.get(MinePatch, keep_patch_id)
    if m:
        db.session.delete(m)
    p = db.session.get(Project, keep_pid)
    if p:
        db.session.delete(p)
    db.session.commit()
print()
print(LINE)
print("六、同一图斑不得重复划分同类型微地质单元")
print(LINE)
# 论文 4.1 节把图斑分为平台、碎石边坡、岩质边坡三类，每类至多一个。
# 允许重复会让面积占比、按类型组织的指标数据与批量导入对不上号。
pid4 = new_patch("D")
c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "1.0"},
       follow_redirects=True)
with app.app_context():
    n1 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n1 == 1, "首次添加「平台」成功", "单元数 %d" % n1)
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "平台",
    "area": "2.0"},
            follow_redirects=True)
with app.app_context():
    n2 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n2 == 1, "重复添加同类型被拒绝", "单元数 %d" % n2)
ok("同一类型只能划分一个" in r.get_data(as_text=True), "给出重复提示")
r = c.post("/unit/%d/add" % pid4, data={"unit_type": "火山口",
    "area": "1.0"},
```

```
            follow_redirects=True)
with app.app_context():
    n3 = MicroUnit.query.filter_by(patch_id=pid4).count()
ok(n3 == 1, "非法单元类型被拒绝", "单元数 %d" % n3)
c.post("/unit/%d/add" % pid4, data={"unit_type": "碎石边坡",
    "area": "3.0"},
        follow_redirects=True)
with app.app_context():
    uid = MicroUnit.query.filter_by(patch_id=pid4,
                                    unit_type="碎石边坡").first(
    ).id
r = c.post("/unit/edit/%d" % uid,
            data={"unit_type": "平台", "area": "3.0"},
    follow_redirects=True)
with app.app_context():
    types = sorted(u.unit_type for u in
                   MicroUnit.query.filter_by(patch_id=pid4)
    all())
ok(types == ["平台", "碎石边坡"], "改类型时同样拦截重复", "、".join(types))
cleanup(pid4, "D")
print()
print(LINE)
if _fails:
    print("存在未通过项：")
    for f in _fails:
        print("  -", f)
    sys.exit(1)
print("流程门禁自检全部通过 √")
print(LINE)
#  ==== 测试/test_weights.py ====
#  -*- coding: utf-8 -*-
"""权重生效链路自检：AHP 算出的权重必须真正参与评价计算。
此前 weight_schemes / weight_items 两张表从未被写入，评价引擎也只用
代码里的论文表 3 默认权重，相当于权重模块算完即弃。本测试守住这条链路：
    默认 → 论文表 3
    保存 AHP 权重 → 评价改用新权重，结果随之变化
    恢复默认 → 结果回到论文取值
"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
#  本脚本位于 测试/ 子目录，先把仓库根目录加入模块搜索路径，
#  使 `from app import ...` 与 app/app.db 的定位不受影响。
ROOT = os.path.dirname(os.path.dirname(os.path.abspath(__file
    __)))
if ROOT not in sys.path:
    sys.path.insert(0, ROOT)
from app import create_app
from app.models import (db, Project, MinePatch, MicroUnit,
    IndicatorData,
```

```
                                ConstraintCheck, EvaluationResult,
    WeightScheme,
                                WeightItem)
from app.services import workflow as wf
from app.services import weight_service as ws
LINE = "=" * 72
BASE = ROOT
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
_fails = []
def ok(flag, label, extra=""):
    print("  %s %-46s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    if not flag:
        _fails.append(label)
TAG = "W"
UNIT_TYPE = "碎石边坡"
def build_case():
    """建一个只含单一微地质单元的图斑，便于观察权重变化对得分的影响。"""
    with app.app_context():
        p = Project(name="权重测试-%s" % TAG, code="WEIGHT-%s"
    % TAG)
        db.session.add(p); db.session.flush()
        patch = MinePatch(project_id=p.id, name="权重测试图斑-%s"
    % TAG)
        db.session.add(patch); db.session.flush()
        db.session.add(MicroUnit(patch_id=patch.id,
    unit_type=UNIT_TYPE,
                                 area=1.0, ratio=1.0))
        db.session.flush()
        u = MicroUnit.query.filter_by(patch_id=patch.id).fi
    st()
        db.session.add(IndicatorData(
            unit_id=u.id, soil_ratio=0.34, thickness=10,
    ph=9.24,
            organic_matter=4.89, coverage=87.92,
    shannon=0.60,
            slope=28.75, aspect="阳坡"))
        db.session.add(ConstraintCheck(
            patch_id=patch.id, policy_pass=True,
    location_pass=True,
```

```
                pollution_pass=True, geo_pass=True,
    farmland_pass=True,
                passed=True, reason="全部通过"))
            db.session.commit()
            return patch.id
def cleanup(pid):
    with app.app_context():
        from app.models import ScoreDetail
        rids = [r.id for r in EvaluationResult.query.filter_
    by(patch_id=pid)]
        if rids:
            ScoreDetail.query.filter(ScoreDetail.result_id.i
    n_(rids)).delete(
                synchronize_session=False)
        EvaluationResult.query.filter_by(patch_id=pid).delet
    e()
        ConstraintCheck.query.filter_by(patch_id=pid).delete
    ()
        uids = [u.id for u in MicroUnit.query.filter_by(pat
    h_id=pid)]
        if uids:
            IndicatorData.query.filter(IndicatorData.unit_id
    .in_(uids)).delete(
                synchronize_session=False)
        MicroUnit.query.filter_by(patch_id=pid).delete()
        m = db.session.get(MinePatch, pid)
        if m:
            db.session.delete(m)
        p = Project.query.filter_by(code="WEIGHT-%s" %
    TAG).first()
        if p:
            db.session.delete(p)
        db.session.commit()
def run_eval(pid):
    c.post("/evaluate/%d" % pid, data={},
    follow_redirects=False)
    with app.app_context():
        return EvaluationResult.query.filter_by(patch_id=pid
    ) \
            .order_by(EvaluationResult.id.desc()).first().fi
    nal_score
print(LINE)
print("一、默认应使用论文表 3 的权重")
print(LINE)
with app.app_context():
    ws.reset_weights(UNIT_TYPE)
    w, label, scheme = ws.effective_weights(UNIT_TYPE)
ok(scheme is None and "论文表" in label, "未配置时权重来源为论文表 3",
    label)
ok(abs(w["l1"]["土壤指标"] - 0.44) < 1e-9, "一级权重取自论文表 3",
```

```
    "土壤指标 %.2f" % w["l1"]["土壤指标"])                                    12203
pid = build_case()                                                          12204
base_score = run_eval(pid)                                                  12205
print("   默认权重下的评价值 F = %.4f" % base_score)                          12206
print()                                                                     12207
print(LINE)                                                                 12208
print("二、保存 AHP 算出的权重后，评价应改用新权重")                          12209
print(LINE)                                                                 12210
with app.app_context():                                                     12211
    # 模拟 AHP 计算结果：把土壤指标权重提高、植被与地形降低（仍和为 1）          12212
    ws.save_weights(UNIT_TYPE, [("soil", 0.70), ("veg",                     12213
    0.20),                                                                  12214
                                ("terrain", 0.10)])                         12215
    n_items = WeightItem.query.count()                                      12216
ok(n_items >= 3, "weight_items 表已写入权重条目", "%d 条" % n_items)          12217
with app.app_context():                                                     12218
    w2, label2, scheme2 = ws.effective_weights(UNIT_TYPE)                   12219
ok(scheme2 is not None and "自定义" in label2, "权重来源变为自定义方案",      12220
    label2)                                                                 12221
ok(abs(w2["l1"]["土壤指标"] - 0.70) < 1e-9, "新权重已生效",                  12222
    "土壤指标 %.2f" % w2["l1"]["土壤指标"])                                  12223
ok(abs(w2["l1"]["植被指标"] - 0.20) < 1e-9, "同批保存的其他项也已生效")       12224
ok(abs(w2["l2"]["土壤指标"]["土石比"] - 0.25) < 1e-9,                        12225
    "未覆盖的二级权重仍保留论文取值",                                        12226
    "土石比 %.2f" % w2["l2"]["土壤指标"]["土石比"])                          12227
new_score = run_eval(pid)                                                   12228
print("   新权重下的评价值 F = %.4f" % new_score)                             12229
ok(abs(new_score - base_score) > 1e-4,                                      12230
    "权重变化确实改变了评价结果",                                            12231
    "%.4f → %.4f" % (base_score, new_score))                                12232
print()                                                                     12233
print(LINE)                                                                 12234
print("三、结果应记录所用权重来源")                                          12235
print(LINE)                                                                 12236
import json                                                                 12237
with app.app_context():                                                     12238
    rec = EvaluationResult.query.filter_by(patch_id=pid) \                  12239
          .order_by(EvaluationResult.id.desc()).first()                     12240
    stored = json.loads(rec.unit_scores)                                    12241
ok(any(u.get("weight_source") for u in stored),                             12242
    "评价记录中标注了权重来源",                                              12243
    stored[0].get("weight_source") or "")                                   12244
print()                                                                     12245
print(LINE)                                                                 12246
print("四、恢复默认后应回到论文取值")                                        12247
print(LINE)                                                                 12248
with app.app_context():                                                     12249
    ws.reset_weights(UNIT_TYPE)                                             12250
    w3, label3, scheme3 = ws.effective_weights(UNIT_TYPE)                   12251
ok(scheme3 is None and "论文表" in label3, "已恢复为论文表 3 权重",          12252
```

```
    label3)                                                          12253
back_score = run_eval(pid)                                           12254
print("   恢复后评价值 F = %.4f（默认时 %.4f）" % (back_score,          12255
    base_score))                                                     12256
ok(abs(back_score - base_score) < 1e-9, "评价结果与默认时一致")        12257
print()                                                              12258
print(LINE)                                                          12259
print("五、权重首页概览")                                             12260
print(LINE)                                                          12261
with app.app_context():                                              12262
    rows = ws.summary()                                              12263
ok(len(rows) == 3, "三种微地质单元均有权重来源说明", "%d 行" % len(rows))  12264
r = c.get("/ahp/")                                                   12265
body = r.get_data(as_text=True)                                      12266
ok(r.status_code == 200 and "当前生效权重" in body, "权重首页正常展示")    12267
print()                                                              12268
print(LINE)                                                          12269
print("六、一致性不通过的矩阵不得保存生效")                            12270
print(LINE)                                                          12271
with app.app_context():                                              12272
    ws.reset_weights(UNIT_TYPE)                                      12273
n = 3                                                                12274
# 明显矛盾的矩阵：a12=9, a13=1/9, a23=9 —— CR 必然超限                12275
mats = [(0, 1, "9"), (0, 2, "0.1111"), (1, 2, "9")]                  12276
data = {"csrf_token": "", "action": "save"}                          12277
for i, j, v in mats:                                                 12278
    data["m_%d_%d" % (i, j)] = v                                     12279
r = c.post("/ahp/matrix?unit_type=%s&level=1" % UNIT_TYPE,           12280
    data=data,                                                       12281
              follow_redirects=True)                                 12282
with app.app_context():                                              12283
    _, label4, scheme4 = ws.effective_weights(UNIT_TYPE)             12284
ok(scheme4 is None, "CR 超限的矩阵未被保存生效", label4)              12285
ok("一致性" in r.get_data(as_text=True), "页面给出了未通过一致性的提示")  12286
cleanup(pid)                                                         12287
print()                                                              12288
print(LINE)                                                          12289
if _fails:                                                           12290
    print("存在未通过项：")                                           12291
    for f in _fails:                                                 12292
        print("  -", f)                                             12293
    sys.exit(1)                                                      12294
print("权重生效链路自检全部通过 √")                                   12295
print(LINE)                                                          12296
# ==== app/services/__init__.py ====                                 12297
# -*- coding: utf-8 -*-                                              12298
# ==== app/utils/__init__.py ====                                    12299
# -*- coding: utf-8 -*-                                              12300
# ==== app/views/__init__.py ====                                    12301
# -*- coding: utf-8 -*-                                              12302
```